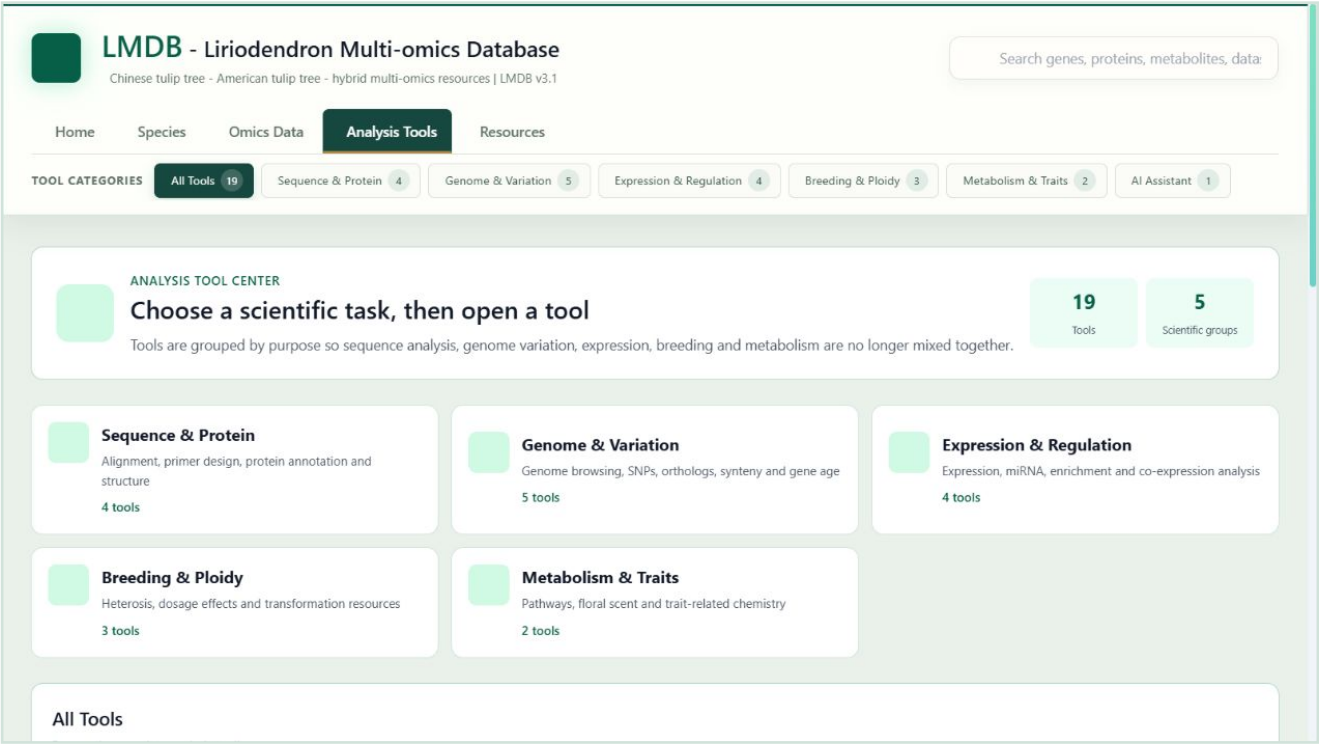
Liriodendron Multi-omics Database Complete User Manual

A practical guide to searching the database and using all 24 listed tools.

This manual documents 19 main-menu tools, two primer-design subtools and three additional pages. It includes real website screenshots for 22 tools. Online checks and screenshots could not be completed for two additional pages; their chapters identify this status. Screenshots preserve the original website interface and may include Chinese labels.

Open the Liriodendron Multi-omics Database



How to use this manual

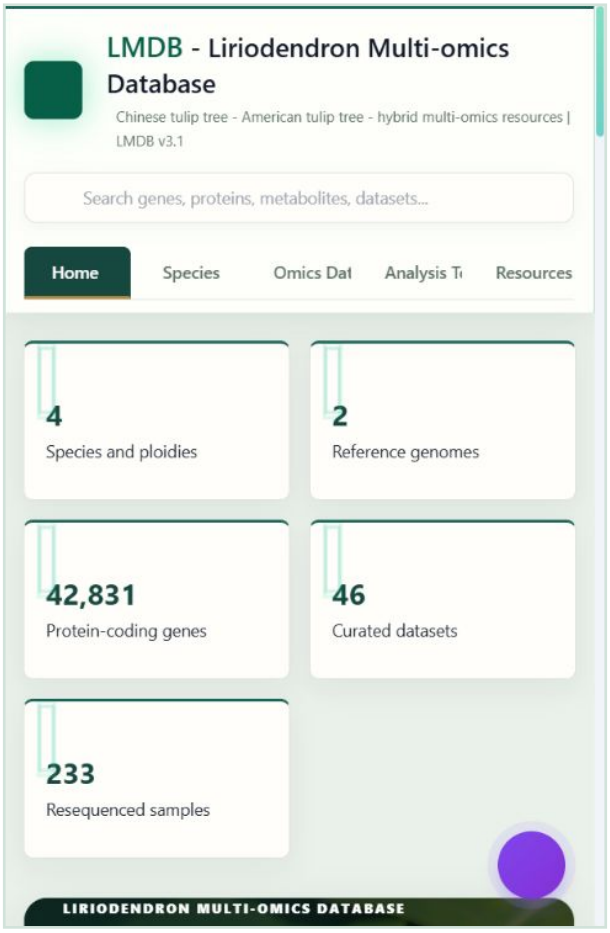
Start with the five quick-start chapters, or use the clickable contents, PDF bookmarks, or tool index to jump to a specific page. Record the species, gene or project ID, data version, parameters and date accessed.

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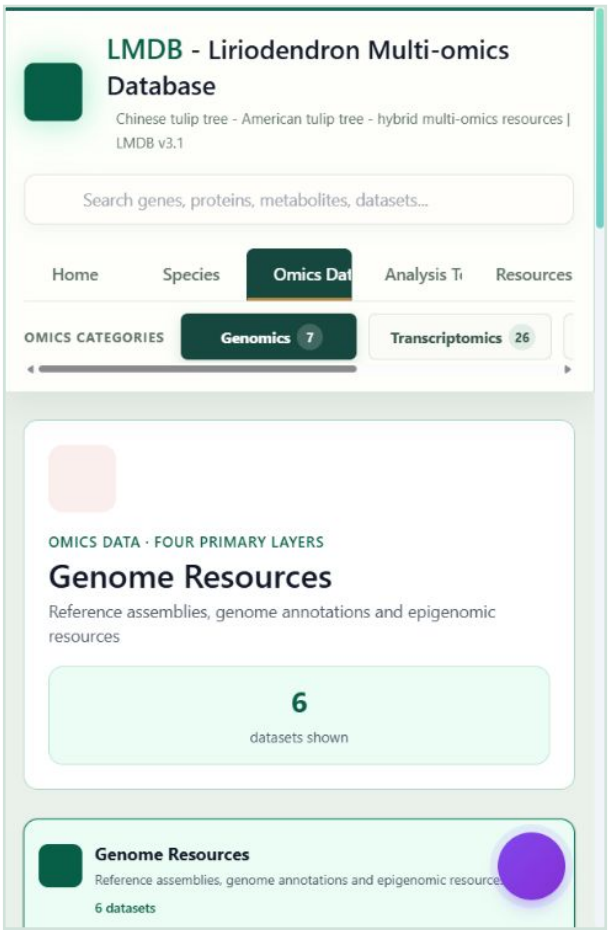
Quick Start 1 Open the website and select the omics module

Enter the data directory from the home page Omics Data. Navigation in narrow screen screenshots slides horizontally, showing more content in wide screen.



Home page and top navigation. Online screenshot: 2026-10-20.

Home page and top navigation. Online screenshot: 2026-10-20.



Omics Classification by Omics Data; click Transcriptomics.

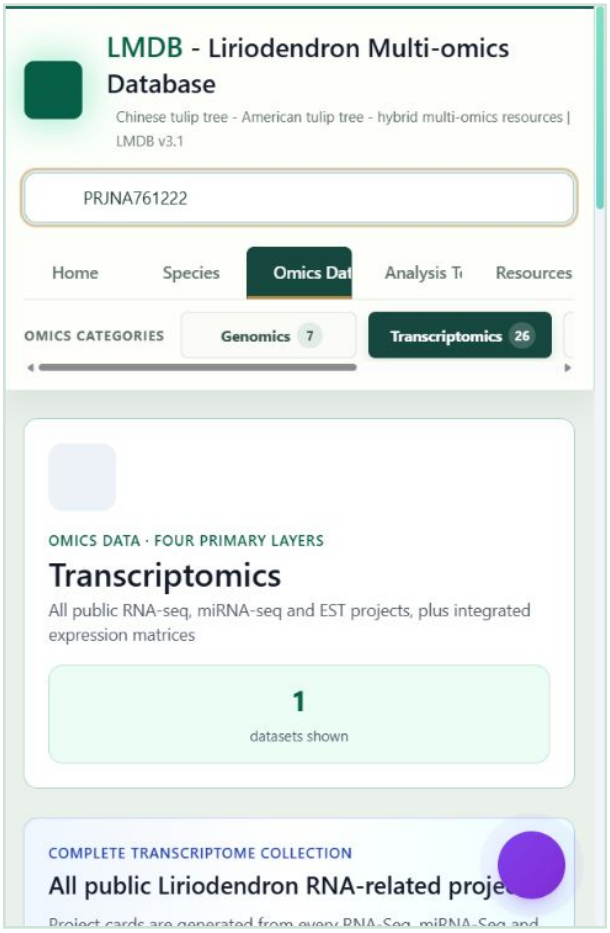
Omics Data; click Transcriptomics.

1. Open the database homepage and click Omics Data.
2. Click Transcriptomics in Omics categories. When looking for genomes, choose Genomics; when looking for protein or metabolism research, choose Proteomics/Metabolomics.
3. The search box at the top filters the project metadata of the current omics and species range in the data directory. To query gene expression, you need to enter Open Expression Atlas. The main menu of

has 19 tools; Primer3 Design also has 2 sub-tools. This manual also contains 3 additional pages for a total of 24 items. Each chapter retains the English control names of the web page to facilitate step-by-step comparison.

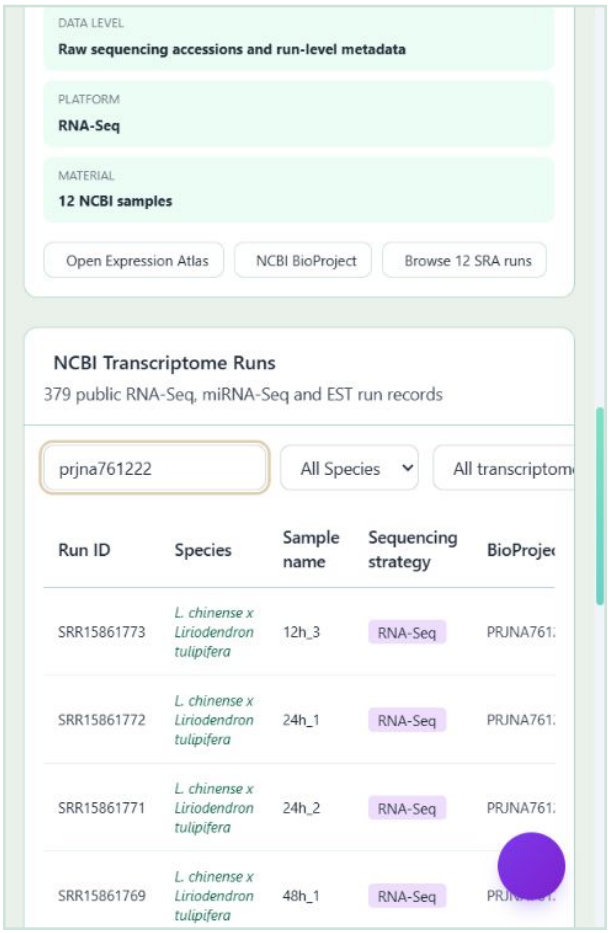
Quick Start 2 Search and check by project number SRA run

sample project PRJNA761222. The project card and NCBI Transcriptome Runs below are two search areas.



Top input PRJNA761222, 1 datasets shown.

top input PRJNA761222, display 1 datasets shown.



Search for run below and enter prjna761222 in lowercase.

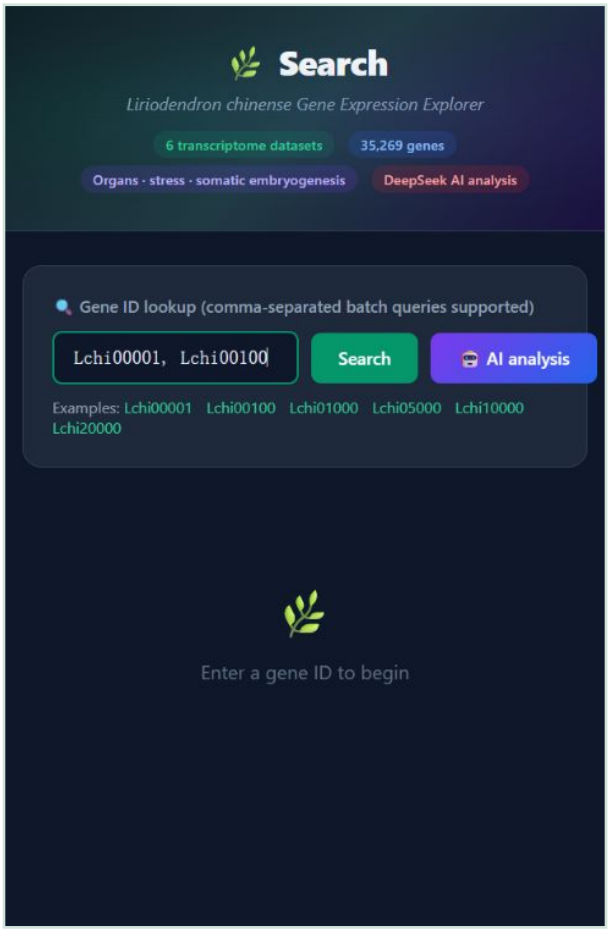
and enter lowercase prjna761222.

1. Enter Transcriptomics, enter PRJNA761222 in the top search box, and the project cards will be filtered instantly.
2. Expand the Cold stress card down and check the project number, RNA-Seq and 12 run/sample records; NCBI BioProject and Browse 12 SRA runs are used for retrospective archiving.
3. In the Search Run ID, BioProject, Sample Name... of NCBI Transcriptome Runs, enter prjna761222 in lowercase; keep All Species / All transcriptome.
4. Check Run ID, Species, Sample name, Sequencing strategy and BioProject. The right column of the table can be viewed by sliding horizontally.

raw sequencing files need to be obtained through the corresponding NCBI/SRA records. Save project number, run ID, species/material and access date; the metadata table on the web page is not equivalent to FASTQ.

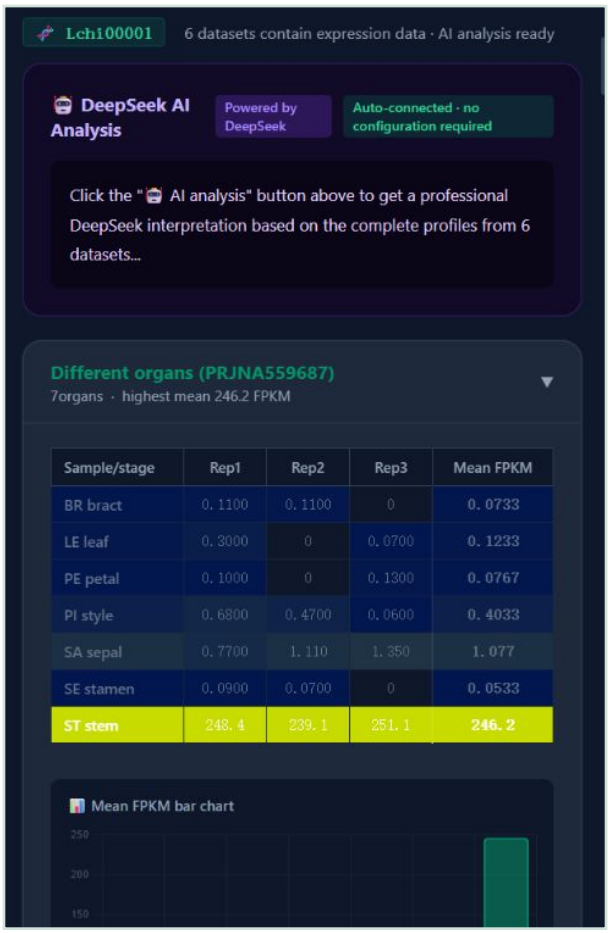
Quick Start 3 Query Gene Expression in Batch

Click Open Expression Atlas and wait for the expression library to load. Example batch values are Lchi00001, Lchi00100.



Gene ID lookup Enter both IDs, click Search or press Enter.

Gene ID lookup Enter both IDs, click Search or press Enter. PRJNA559687 organ expression table for



Lchi00001.

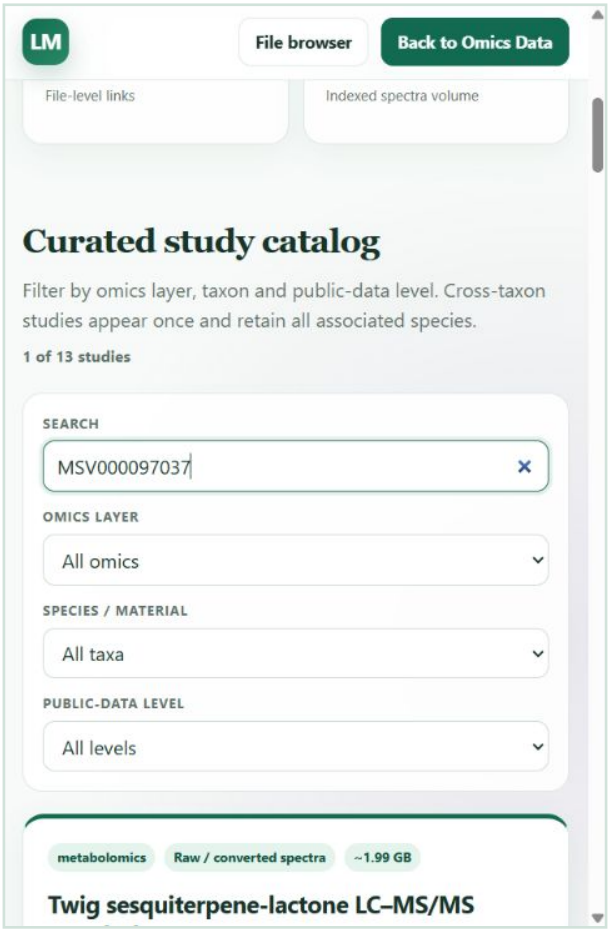
Lchi00001.

1. Enter Lchi00001, Lchi00100 in Gene ID lookup; English commas, Chinese commas or blanks can be separated. Click Search.
2. Results are grouped by gene. Check the gene ID and each dataset title; click on the title to collapse it, and continue scrolling down to view the second gene.
3. Sample/stage is the tissue/stage, Rep1-3 is the repetition, and Mean FPKM is the mean. The ST stem in this example is 246.2 FPKM. The

expression page displays an FPKM table, bar chart, and line chart without differential expression testing or P/FDR. AI analysis recommends querying a gene individually before use. See the Expression Atlas chapter for complete operations.

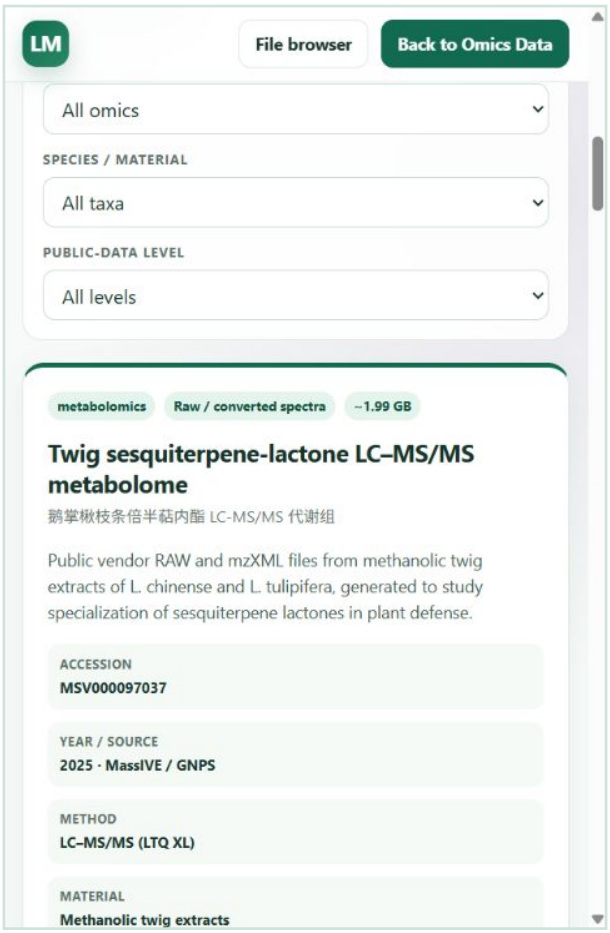
Quick Start 4 Search Proteome/Metabolome Studies

Open the Curated study catalog, search for public projects by research metadata, and then go to the file catalog.



Search input MSV000097037; results are 1 of 13 studies.

Search input MSV000097037; result is 1 of 13 studies.



Check the species, method and Accession of the research card.

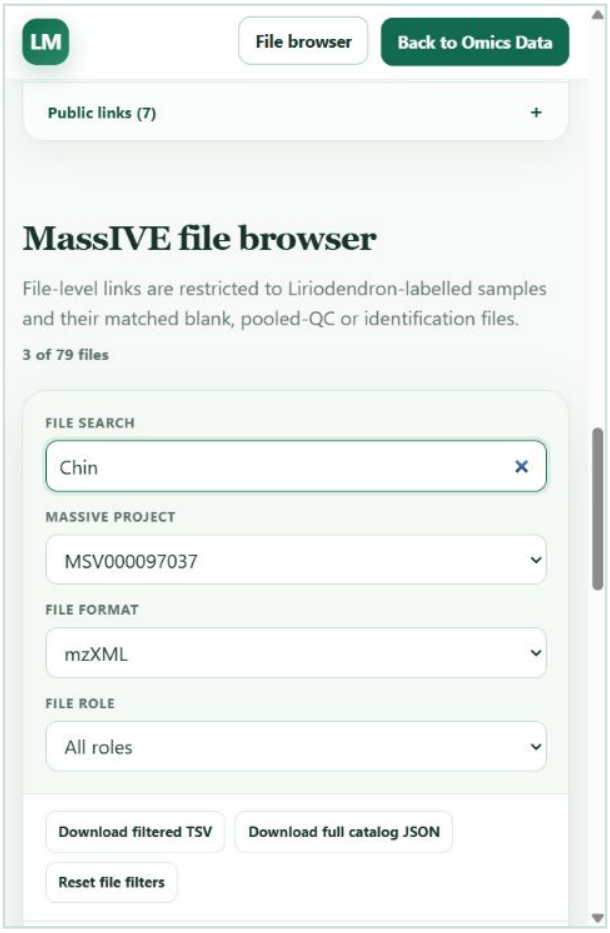
Check the species, method and Accession of the research card.

1. Enter MSV000097037 in Search and keep All omics / All taxa / All levels.
2. Check Twig sesquiterpene-lactone LC-MS/MS metabolome and its Accession. You can use Omics layer, Species/material, and Public-data level to narrow the scope of your research.
3. Click File browser at the top to enter the file area. Public links expand papers and public archive entries.

research search targets metadata such as title, accession number, methods, organization, etc. Publication only records may only have paper information; confirm the data level such as Raw / converted spectra or Processed public data before obtaining the data.

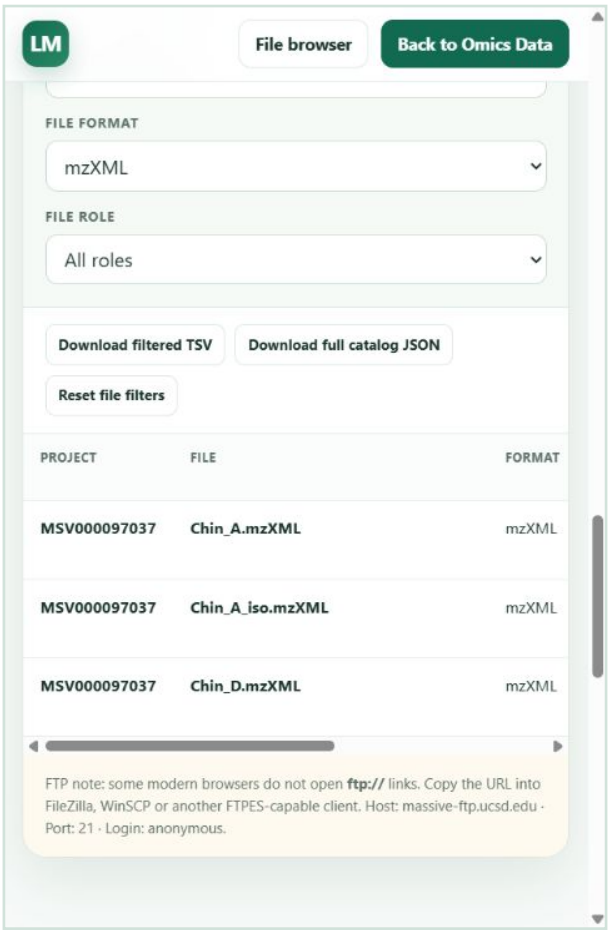
Quick Start 5 Filter files and save download list

Select the project, file keywords and format in the MassIVE file browser. Example result is 3 of 79 files.



MSV000097037 + File search = Chin + File format = mzXML□

MSV000097037 + File search = Chin + File format = mzXML□



Matches Chin_A.mzXML, Chin_A_iso.mzXML, Chin_D.mzXML.

matches Chin_A.mzXML, Chin_A_iso.mzXML, Chin_D.mzXML.

1. Select MSV000097037 for MassIVE project, enter Chin for File search, select mzXML for File format, and keep All roles for File role.
2. View three file records; Open opens the external file address, and Copy copies the link. Slide the document table sideways when needed.
3. Download filtered TSV saves the current file list and address, Download full catalog JSON saves the complete catalog. The original spectra are available separately via links to each file. The

FTP link can be obtained from a compatible client or from the public project page. When the result is empty, click Reset file filters and re-add the condition. TSV/JSON exports directories and does not download all original data in batches.

NCBI BLAST · Sequence similarity search

sequence and protein | Calling external NCBI services

Entrance

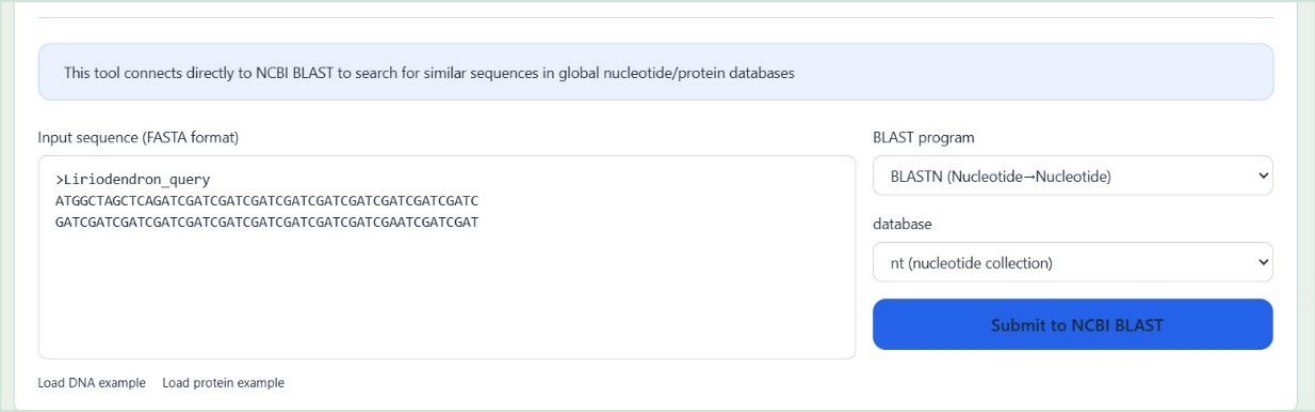
Home Page → Analysis Tools → Sequence & Protein → NCBI BLAST (tool key blast)

Open website/independent page

Function

Submit nucleic acid or protein sequences to the NCBI public BLAST service to find similar sequences in public databases.

2026-10-21 Screenshot after opening and filling in the built-in examples on the page. No NCBI comparison results were obtained in this round; the following submission and result descriptions are based on the operation logic of the current page.



NCBI BLAST · Sequence similarity search

real online screenshot: 2026-10-21.

Input and parameters

- Input sequence (FASTA format): Paste a FASTA or pure sequence; remove the FASTA header and blanks and leave at least 10 characters.
- Load DNA example / Load protein example: Fill in the built-in demonstration sequence on the page; only responsible for filling in, not automatically submitted.
- BLAST program□BLASTN (Nucleotide→Nucleotide)□BLASTP (protein to protein)□BLASTX (nucleotide to protein)□TBLASTN (protein to nucleotide)□
- database: nt (nucleotide collection), nr (protein collection), RefSeq RNA, RefSeq Protein, SwissProt. Default blastn + nt.

Sample process The

page actually provides Load DNA example and Load protein example. You can choose Load DNA example → BLASTN → nt → Submit to NCBI BLAST; this is a demonstration operation path, and there is no guarantee that the demonstration sequence will have meaningful biological hits.

Step by step operation

1. Paste your own target sequence, or use Load DNA example / Load protein example to familiarize yourself with the input method. Please replace the real analysis with the verified target sequence.
2. Select the BLAST program and compatible database according to the query type: BLASTN + nt for nucleic acid query; BLASTP + nr or SwissProt for protein query.
3. Click Submit to NCBI BLAST; after submission, the page displays Processing at NCBI....
4. If the Job ID is obtained, save the RID and click View full results at NCBI to view the complete hits and alignments.
5. If the page gets parsable BLASTOutput2, up to 20 hits will be displayed; use the RID to query NCBI when the network fails or polling times out.

How to read the result

- The hit card on this page displays the title, Accession, Score (bit score), E-value, Identity (identity/align_len percentage of the first HSP).
- E-value reflects the expected number of random occurrences of this type of alignment under the database and parameters; identity is the consistent proportion within the local alignment, which needs to be combined with complete alignment coverage and species judgment.
- View full results at NCBI is used to confirm sequence coverage, alignment intervals and more hits; this page does not display the full sequence alignment.

Frequently Asked Questions and Usage Restrictions

- This is an online query for the NCBI global public library, not the database's local Liriodendron tulipifera reference sequence library.
- The local form does not have E-value threshold, maximum number of hits, or species qualification controls, and these parameters should not be fictitious in the manual.
- The request relies on NCBI services, network and browser cross-domain access; after submission, the actual task status and RID shall prevail.
- The program and database will not be automatically paired, and users need to select a compatible combination. The built-in sequences are demo inputs and cannot be used as real study sequences.

Primer3 Design · Primer design portal

sequence and protein | External primer service and sub-tool entrance of this site

Entrance

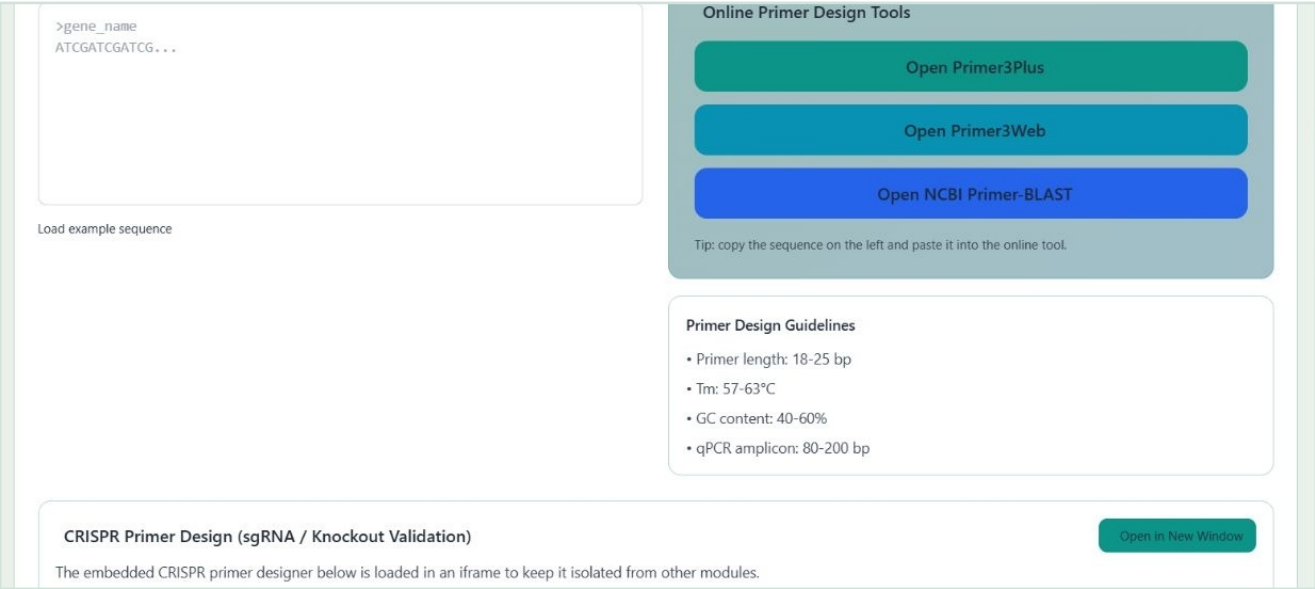
Homepage → Analysis Tools → Sequence & Protein → Primer3 Design (tool key primer); CRISPR and overexpression primer sub-tools are also embedded at the bottom of the same page.

Open website/independent page

Function

provides sequence staging area, external Primer3 / Primer-BLAST entrance, and brings together two local primer design subpages.

2026-10-21 has been opened from the online address and taken a screenshot.



Primer3 Design · Primer design entrance

real online screenshot: 2026-10-21.

Input and parameters

- Input template sequence (copy to an online tool): Paste the template DNA sequence.
- Load example sequence: Fill in the FASTA demonstration sequence built into the page.
- External entry buttons: Open Primer3Plus, Open Primer3Web, Open NCBI Primer-BLAST.

Sample process

Load example sequence → Manually copy FASTA → Open Primer3Plus → Paste to external website. Real experiments need to be replaced with their own target templates.

Step by step operation

1. Paste your own template sequence into Input template sequence (copy to an online tool).
2. Copy the sequence on the left and click Open Primer3Plus, Open Primer3Web or Open NCBI Primer-BLAST.
3. Paste the sequence in the external page that opens, and set parameters such as product length and primer length according to the actual interface of the service before running.
4. When you need sgRNA and knockout validation primers, use CRISPR Primer Design (sgRNA / Knockout Validation) at the bottom of this page.
5. When you need overexpression primers with vector homology arms, use Gene Overexpression Primer Design (Vector / Homology Arms / Tags); click Open in New Window.

How to read the result

- The top sequence box itself does not generate Primer3 results, nor does it automatically pass the sequence to an external website; must be copied and pasted.
- The calculations and results of common PCR primers can be viewed on the external website. The length 18-25 bp, Tm 57-63°C, GC 40-60%, and qPCR product 80-200 bp given on the page are recommended text, not parameter controls on this page.
- The local results and export methods of the two embedded sub-tools are described in their respective chapters.

Frequently Asked Questions and Usage Restrictions

- Do not describe this top-level module as calling the Primer3 engine on this site to complete conventional PCR design. The current page implementation only provides external links here.
- External website parameters, results and availability are controlled by the corresponding services; this manual does not write external forms that have not been accessed and verified as controls on this site.
- Primer candidates still need to check the target sequence, amplification specificity and experimental suitability.

CRISPR Primer Designer · sgRNA and validation primers

sequence and protein | Browser local rule calculation

Entrance

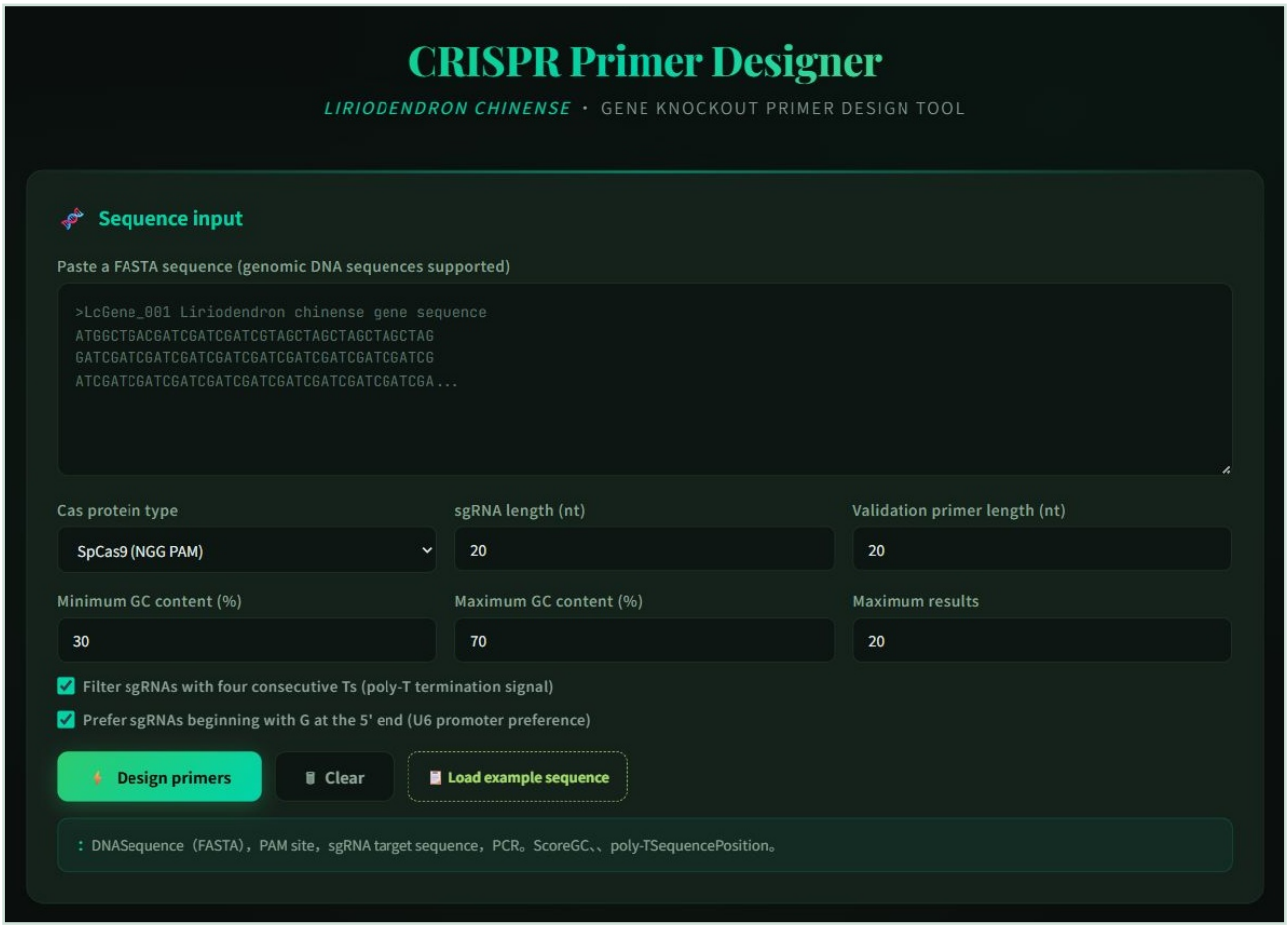
Analysis Tools → Sequence & Protein → Primer3 Design → CRISPR Primer Design; you can also Open in New Window, or open crispr_primer_designer.html directly.

Open website/independent page

Function

scans PAM in the browser based on the DNA pasted by the user, generates candidate sgRNA and gives surrounding validation PCR primers.

2026-10-21 has been opened from the online address and taken a screenshot.



CRISPR Primer Designer · sgRNA and verification primers

real online screenshot: 2026-10-21.

Input and parameters

- Paste a FASTA sequence (genomic DNA sequences supported): seqInput, at least 50 bp; one sequence at a time is recommended.
- Cas protein type□SpCas9 (NGG PAM)□SaCas9 (NNGRRT PAM)□Cas12a/Cpf1 (TTTV PAM)□
- sgRNA length (nt): Default 20, input box range 17-25. Validation primer length (nt): Default 20, range 18-30.
- Minimum GC content (%) / Maximum GC content (%): Default 30 / 70. Maximum results: Default is 20, input box range is 5-50.
- Filter sgRNAs with four consecutive Ts (poly-T termination signal): checked by default. Prefer sgRNAs beginning with G at the 5' end (U6 promoter preference): checked by default.

Sample process

Click Load example sequence (actual built-in title LcAG / AGAMOUS-like example fragment), keep SpCas9, 20nt, GC30-70%, Maximum results20, and then click Design primers. The example is used to familiarize yourself with the operation; for real analysis, please replace it with your own target sequence.

Step by step operation

1. Paste the DNA containing the target region and sufficient flanking areas on both sides, or click Load example sequence to load the LcAG demonstration fragment from the web page.
2. Select Cas protein and PAM, adjust sgRNA length, verification primer length, GC upper and lower limits, and maximum number of results; retain poly-T filtering and 5'G preference as needed.
3. Click Design primers. Automatically scroll to the results area after the results are completed.
4. Check the position, strand, sgRNA, PAM, GC% and Score at sgRNA target sites.
5. Switch to Validation primers, click on the candidate primer card to expand and view the forward and reverse sequences, Tm, GC, coordinates, and Expected product / Product size.
6. Switch Sequence view to check the position of the target and PAM in the input sequence; you can Copy, Copy primer pair, Export CSV or Copy all primers.

How to read the result

- Score is the tool's own 0-100 heuristic ranking: primarily considering GC, self-complementarity, poly-T, starting G, and position in the input sequence; not equivalent to experimental editing efficiency.
- Positions are displayed as input sequence coordinates and are not automatically mapped to chromosome coordinates. sgRNA results are associated with the corresponding validation primers with the same target number.
- The verification primer sequence is output in 5′→3′, and Reverse has been reverse complementary; CSV also contains sgRNA, PAM, GC, score, forward and reverse primers, and product size.
- The statistics card gives the input sequence length, number of candidates, average GC and average score; the results are first sorted in descending order by local score, and then the Maximum results are intercepted.
- Score is a heuristic score of the page based on a combination of rules such as GC, self-complementary, poly-T, 5'G and position, and is not an experimentally calibrated editing efficiency or off-target probability.
- The verification primer takes the specified length from about 150 bp upstream and downstream of the candidate cut point, and then reports the expected product and empirical Tm.

Frequently Asked Questions and Usage Restrictions

- Only scans the input sequence and does not perform whole-genome off-target, exon annotation or reference variant checking; Score is a heuristic sorting and cannot be regarded as an experimentally calibrated editing efficiency.
- Enter one target sequence at a time. Multiple FASTAs will be spliced and not designed separately; retaining sufficient sequence on both sides of the target will help generate verification primers.
- Validation primers are ordered at a fixed offset near the target, Tm is a simplified estimate; there is no genome-wide specificity, hairpin, or primer-dimer optimization.
- All Cas types use the same approximate cut point logic. SaCas9/Cas12a cut points, negative-strand candidate positions, plotted coordinates, and predicted products should all be reviewed against the actual reference sequence.

Overexpression Primer Designer · Gene overexpression primers

sequence and protein | Browser local sequence calculation

Entrance

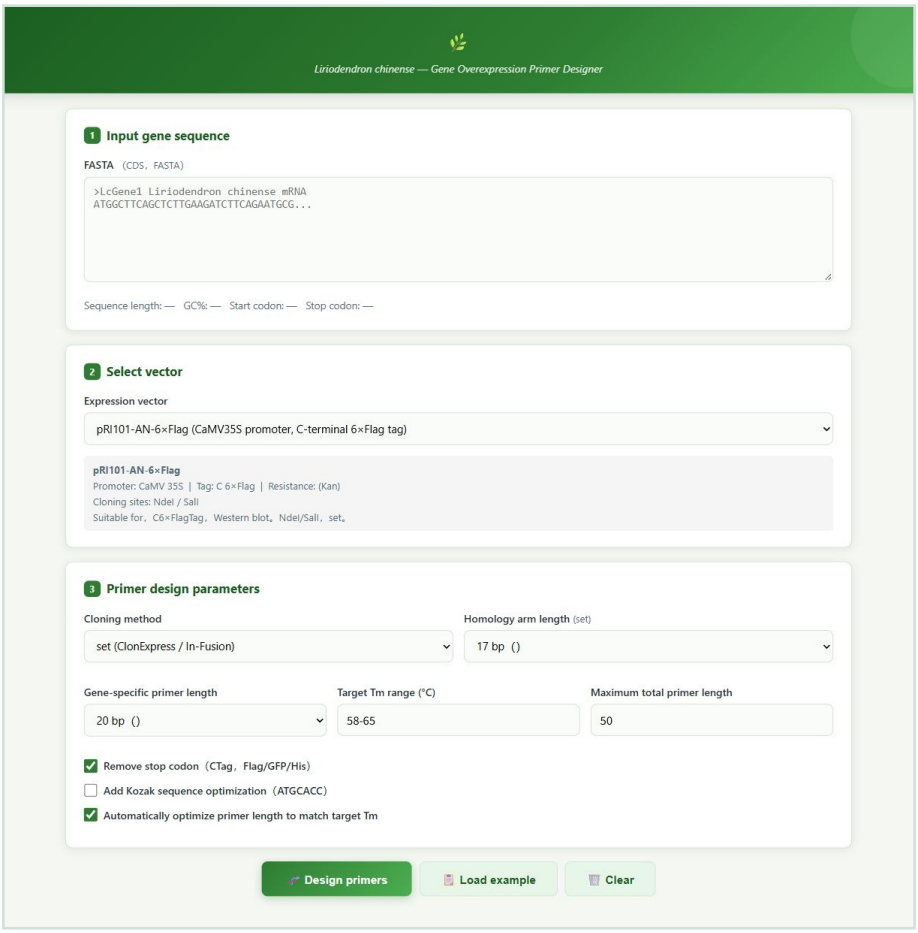
Analysis Tools → Sequence & Protein → Primer3 Design → Gene Overexpression Primer Design; available Open in New Window, or open primer_designer.html directly.

Open website/independent page

Function

generates overexpression amplification primers with homology arms or enzyme digestion sequences in the browser based on the sequences at both ends of a CDS and vector settings.

2026-10-21 has been opened from the online address and taken a screenshot.



Overexpression Primer Designer · Overexpression primer

real online screenshot: 2026-10-21.

Input and parameters

- FASTA sequence (paste a CDS sequence; FASTA header optional): fastaInput, at least 30 bp; if it does not start with ATG, you will be prompted for confirmation.
- Overexpression vector□pRI101-AN-6×Flag□pRI101-AN-GFP□pCAMBIA1300-35S□pCAMBIA2300-35S□pBI121□pSuper1300-GFP□pCAMBIA1305-GFP□pHB-Flag□pGreenII 62-SK□Custom vector□
- Cloning method (design parameter area): Homologous recombination (ClonExpress / In-Fusion) or Restriction digest and ligation.
- Homology arm length: 15/17 (default)/20/25 bp. Gene-specific primer length: 18/20 (default)/22/24/25 bp.

- Target Tm range: Default 58-65°C. Maximum total primer length: default 50.
- Remove stop codon: checked by default, the terminal stop codon is actually only removed when the carrier hasTag=true. Add Kozak sequence optimization: Not checked by default, you can add CACC before ATG. Automatically optimize primer length to match target Tm: checked by default.
- Custom vector also has Custom vector name, Forward homology arm / upstream sequence of the restriction site, Reverse homology arm / downstream sequence of the restriction site, Vector has a C-terminal tag; the reverse arm automatically reverse complements according to the input instructions.

Sample process

Load example → Keep pRI101-AN-6×Flag, homologous recombination, 17bp arm, 20bp specific region, Tm58-65, Remove stop codon and automatic optimization → Design primers. The source of the example is LcMYB1 built into the web page, which does not replace the real CDS and carrier verified by the user.

Step by step operation

1. Paste the complete target CDS with the correct direction, or use Load example to load the webpage LcMYB1 example.
2. Select the actual vector used, read the label, cloning site and vector prompt in Vector information; fill in the vector name, forward and reverse adapters and C-terminal label options if you need to customize it.
3. Set the Cloning method, homology arm length, specific region length, target Tm and maximum total length in the Primer design parameters area.
4. Press whether to do C-side integration to confirm Remove stop codon, and select Kozak and automatic optimization.
5. Click Design primers; check the linker region and Gene-specific region, stop codon treatment and all warnings in the forward and reverse complete primers.
6. Use Copy sequence, Copy all primers, Export CSV or Export order format to save the sequence; double-check the connection boundaries and reading frames between the actual linearized vector and the inserted fragment before placing an order.

How to read the result

- Outputs two primers, the name consists of the first word of FASTA plus -OE-F/-OE-R, and the complete sequence is provided in 5′→3′ order.
- Tm (specific region) is used to view the first amplification annealing part; the full-length Tm also includes the 5′ addition region, and the two have different meanings.
- The maximum total primer length falls within the result warning threshold and does not force a truncated design.
- An attempt will be made to remove the standard stop codon at the end of the CDS only if Remove stop codon is checked and the vector has a C-terminal tag.
- Output two candidates, the primer names are gene names -OE-F and -OE-R. Display the complete primer, adapter/homologous arm, gene specific region, length, specific region Tm, full-length Tm and GC.
- The summary table includes Primer name, Sequence (5'→3'), Length, Tm (specific region), GC%; prompts that Tm is out of range, GC is inappropriate, total length is out of limit, forward and reverse Tm differences, etc.
- The maximum total primer length is used for result alarms. The current page implementation will not automatically truncate to the maximum length accordingly.

Frequently Asked Questions and Usage Restrictions

- Use the two-terminal primer generation and Tm estimation algorithm on this page, without calling the Primer3 engine, and without doing genome-wide specificity, internal restriction sites or primer-dimer scanning.
- Vector preset adapters must be checked one by one with the actual sequence, linearized position and reading frame of the vector on hand, and cannot be applied just by the vector name.
- One unambiguous CDS is submitted each time. Multiple FASTAs will be concatenated; non-ATCGU characters will be removed and U will be converted to T.
- The Cloning method drop-down box in the custom carrier area does not participate in the final design; the Cloning method in the Primer design parameters area shall prevail.

- Custom restriction enzyme cutting mode will not automatically use hand-filled homology arms as restriction sites. Custom vectors preferentially operate according to the homologous recombination path and check the output; do not assume that the program has added the correct enzyme digestion adapter.
- Maximum total primer length is the result warning threshold and will not be forced to be truncated; when Tm or length does not meet the requirements, the parameters need to be adjusted and redesigned.

UniProt Database · Protein search

sequence and protein | Calling external UniProt services

Entrance

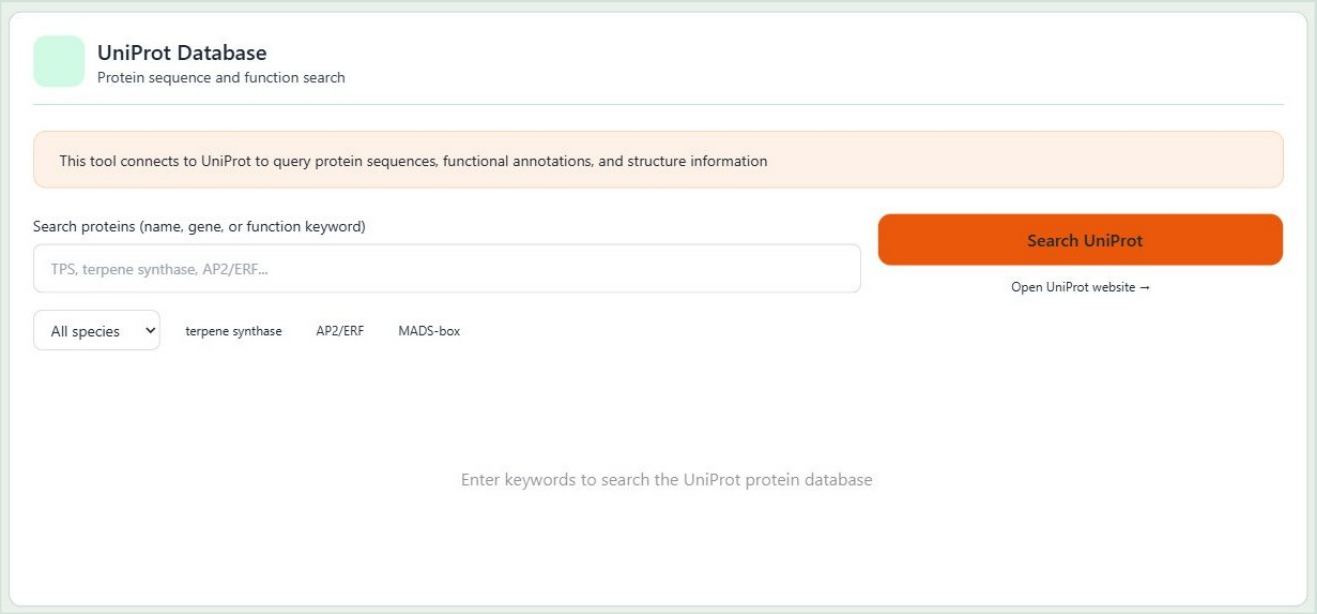
Home Page → Analysis Tools → Sequence & Protein → UniProt Database (tool key uniprot)

Open website/independent page

Function

queries UniProt protein entries, species, and function names, and opens the FASTA and AlphaFold portals.

2026-10-21 has been opened from the online address and taken a screenshot.



UniProt Database · Protein Query

real online screenshot: 2026-10-21.

Input and parameters

- Search proteins (name, gene, or function keyword): Enter the protein name, gene name, or function keyword. The placeholder is TPS, terpene synthase, AP2/ERF....
- Species drop-down: All species, Arabidopsis, rice (Oryza), poplar (Populus), Magnolia. There is no dedicated dropdown item for Liriodendron.
- Shortcut word buttons: terpene synthase, AP2/ERF, MADS-box; only fill in keywords.

Sample process

Click terpene synthase → All species → Search UniProt; after submission, the actual returned results or error prompts shall prevail.

Step by step operation

1. Enter keywords, such as terpene synthase, or click on the corresponding shortcut word.
2. Optional species range. When searching for Liriodendron tulipifera, you can first retain All species and use appropriate UniProt query expressions in keywords.
3. Click Search UniProt or press Enter in the keyword input box.
4. Read entry name, primary accession, species, gene and Swiss-Prot/TrEMBL markers.
5. Click Details to view the complete UniProt record; click FASTA to obtain the sequence, and AlphaFold to open the corresponding structure record; use the Open UniProt website when more results are needed.

How to read the result

- Gets up to 20 entries and displays Found N results. This number is the number of entries returned, not the complete total number of hits for the public database.

-Swiss-Prot tag corresponds to the reviewed entry, and TrEMBL corresponds to the unreviewed type; the annotation and protein name still need to be judged based on the source.

- AlphaFold links are spliced by accession, which does not mean that every hit has a usable structure.

Frequently Asked Questions and Usage Restrictions

- Relies on UniProt REST API and network; not native protein database or site annotation calculations.
- There are no sequence similarity alignment, bulk FASTA upload, or results download buttons on this page; FASTA is an external link for each entry.
- Species drop-down filter by spelling the genus name into organization_name. If there are no results from the filter, return to All species first.

Protein Structure (ESMFold) · Protein structure prediction

sequence and protein | External prediction service and local structure display

Entrance

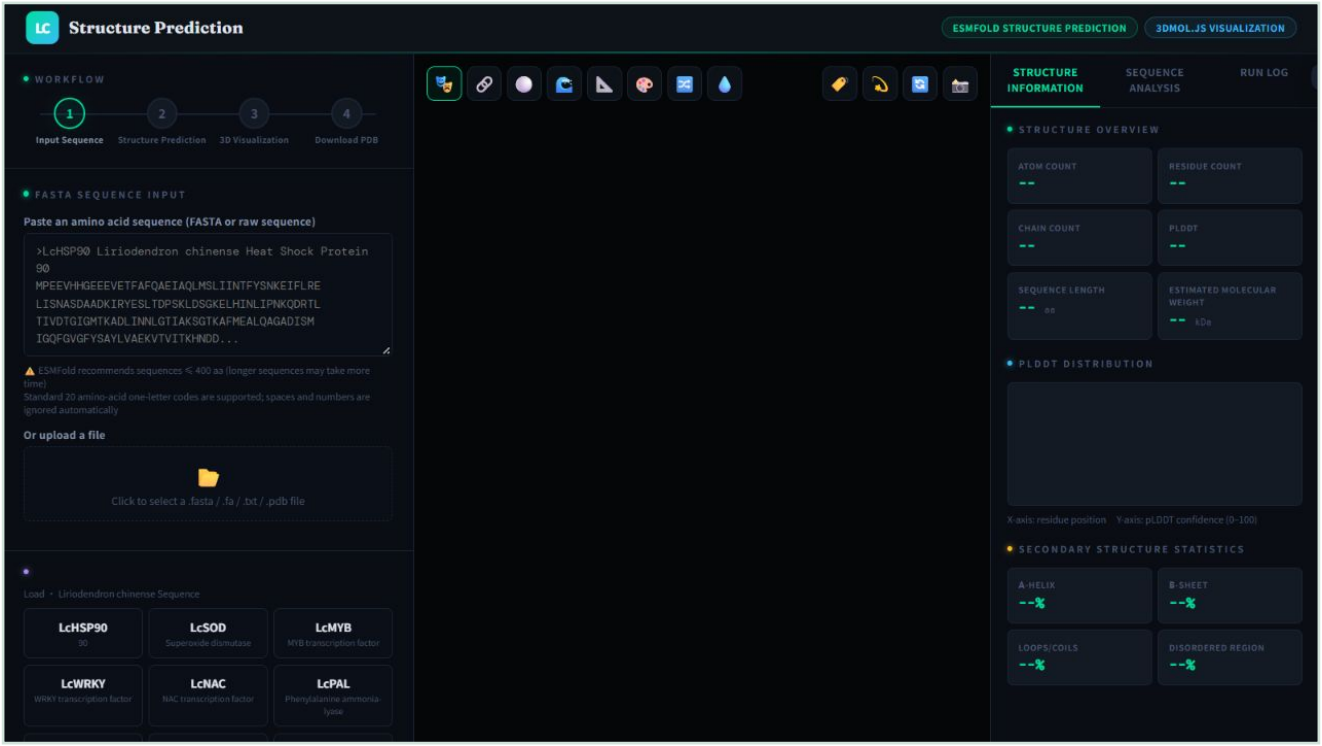
Homepage → Analysis Tools → Sequence & Protein → Protein Structure (ESMFold) (tool key structure); embed independent page assets/index-embedded-01.html.

Open website/independent page

Function

submits the amino acid sequence to the external ESMFold service and returns the PDB to view the structure in 3Dmol.js; you can also import an existing PDB for browsing.

2026-10-21 The online structure tool has been opened and a screenshot has been taken. This round shows the input and structure display area, and no ESMFold prediction has been submitted; whether the external prediction is completed shall be subject to the actual operation log.



Protein Structure (ESMFold) · Protein Structure

real online screenshot: 2026-10-21.

Input and parameters

- Paste an amino acid sequence (FASTA or raw sequence): fasta-input, at least 10 residues; page recommendation ≤400aa.
- Or upload a file: FASTA/FA/FAA/TXT for sequences, PDB/ENT for existing structures. The file box also accepts CIF, but the current parsing implementation is still handled as PDB and cannot guarantee CIF compatibility.
- Liriodendron Example Proteins□LcHSP90□LcSOD□LcMYB□LcWRKY□LcNAC□LcPAL□Lc4CL□LcCHS□LcAPX□

Sample process

Click on the actual example LcSOD → Check sequence length → Predict 3D protein structure (ESMFold). The example is only used as an operation entry point. Please refer to the running log for prediction of completion.

Step by step operation

1. Paste a standard amino acid sequence, upload the sequence file, or click LcSOD and other examples to load; check the Residue count.

2. Click Predict 3D protein structure (ESMFold) and wait for the external service to return; check the Run log and error prompts.
3. After the prediction is completed, rotate and zoom in the 3D window; the display can be switched between Cartoon, Sticks, Spheres, Molecular surface, and Lines.
4. Switch coloring with Color by pLDDT, Color by chain, or Color by hydrophobicity, and view structural information, residue confidence curves, and sequence analysis labels.
5. Use Residue labels, Auto-spin, Reset view to adjust; Save screenshot to save the view; Download PDB structure file to export PDB.
6. If you already have a PDB, you can directly upload it for browsing without submitting a new prediction task.

How to read the result

- Displays the number of atoms, number of residues, number of chains, source, mean pLDDT and residue-by-residue pLDDT curves; the B-factor field of the ESMFold predicted PDB is interpreted as pLDDT.
- Color legend: >90 Very high, 70-90 High, 50-70 Low, <50 Very low; low confidence areas serve as structural uncertainty cues.
- The PDB download name is header_ESMFold.pdb or protein_ESMFold.pdb. Sequence molecular weight and pi are approximate estimates.

Frequently Asked Questions and Usage Restrictions

- The interface is locally embedded, and the structure prediction is actually sent to https://api.esmatlas.com/foldSequence/v1/pdb/; it relies on external services and is not guaranteed to be available immediately.
- The parser will assemble multiple FASTA sequences into one sequence; each time a protein is input, it uses the standard 20 amino acids and checks non-standard residues by itself. Some built-in examples contain non-standard characters such as B and should not be used as examples to guarantee success.
- When importing an existing experimental PDB, the B-factor is usually a temperature factor and cannot be directly interpreted as pLDDT.
- The so-called secondary structure percentage on the right is actually counted according to the pLDDT interval, and is not a DSSP or structural geometry determination; it cannot be used as a scientific conclusion on the α-helix/β-sheet ratio.
- Although it supports selecting the .cif extension, showPDB is uniformly sent to the viewer in 'pdb' format and reads the PDB ATOM column. Therefore, the manual gives priority to instructing PDB and does not guarantee the correct CIF structure.

Genome Browser · Genome region browser

genome and variation | Preset fragment record browsing

Entrance

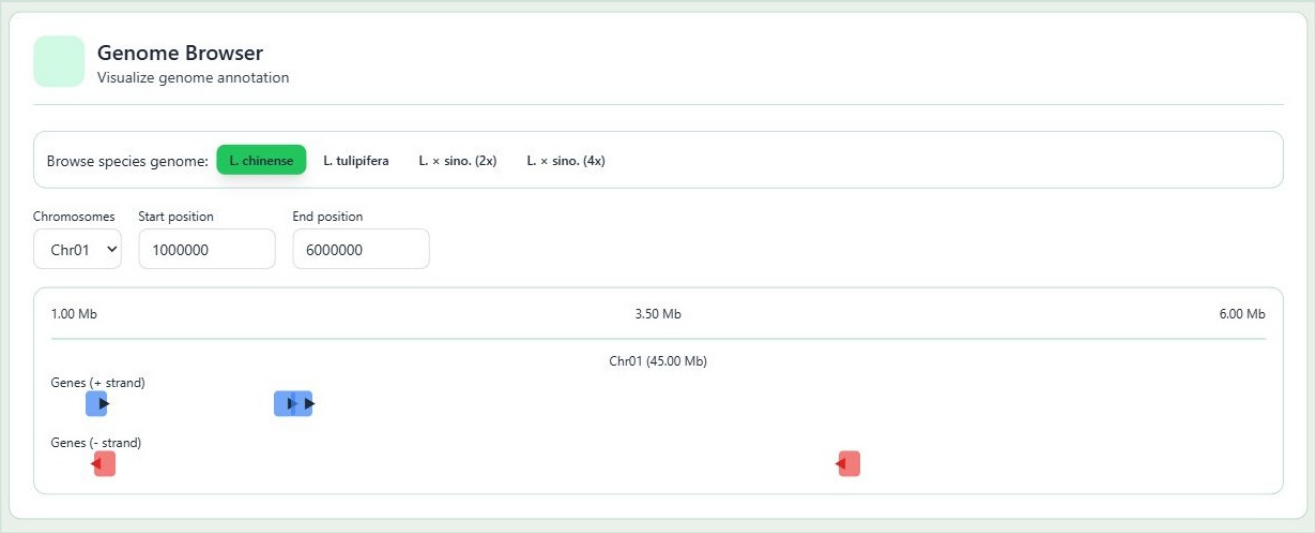
Home Page → Analysis Tools → Genome & Variation → Genome Browser (tool key browser)

Open website/independent page

Function

Browse the gene position and positive and negative strands of the preset chromosome segments on the current page, and click on the gene to view details.

2026-10-21 has been opened from the online address and taken a screenshot.



Genome Browser · Genome fragment browsing

real online screenshot: 2026-10-21.

Input and parameters

- Browse species genome□L. chinense□L. tulipifera□L. × sino. (2x)□L. × sino. (4x)□
- Chromosomes: Only the current preset species keys are listed; Liriodendron tulipifera Chr01-Chr05, and the remaining three categories Chr01-Chr03.
- Start position / End position: bp coordinate number, default 1,000,000 and 6,000,000.
- Four icon buttons: pan left, zoom in, zoom out, pan right, no text submit button.

Sample process

select L. chinense / Chr01, Start position=1000000, End position=6000000, and click Lchi00001 and other gene strips in the preset record. This example uses the preset records included in the page.

Step by step operation

1. Select species, then Chromosomes.
2. Enter the Start position and End position, ensuring that the starting point is smaller than the end point and is within the current chromosome range; the page is updated with the input.
3. Use the zoom in/out icon to change the window, and use the left and right icons to move about 30% of the current window.
4. Click on the visible gene bar in Genes (+ strand) or Genes (- strand).
5. View the gene ID, name, position, chain direction and length in detail below.

How to read the result

- The blue bar is the positive strand and the red bar is the negative strand; the gene bar and window scale are based on the current preset coordinates.
- Details give the start and end coordinates and length of the selected gene; the window and gene positioning are not the results of sequence alignment.

Frequently Asked Questions and Usage Restrictions

- Currently only a small number of chromosome fragments and gene records are included, and all 19 chromosome annotations are not fully loaded. When formal gene positions or structures are required, check corresponding reference annotations or use the additional Gene Structure Viewer.
- There are no gene ID search, FASTA/GFF upload, sequence track or genome download buttons in this interface.
- Before inputting, confirm that the start and end coordinates are valid numbers, and the starting point is smaller than the end point; the wrong range may cause display abnormalities.
- The coordinate/annotation source of the preset data and the official reference version need to be verified separately, and the sample strips should not be directly regarded as scientific research evidence.

Population SNP Atlas · Population variant browser

genome and variation | Precalculated population result browsing and table import

Entrance

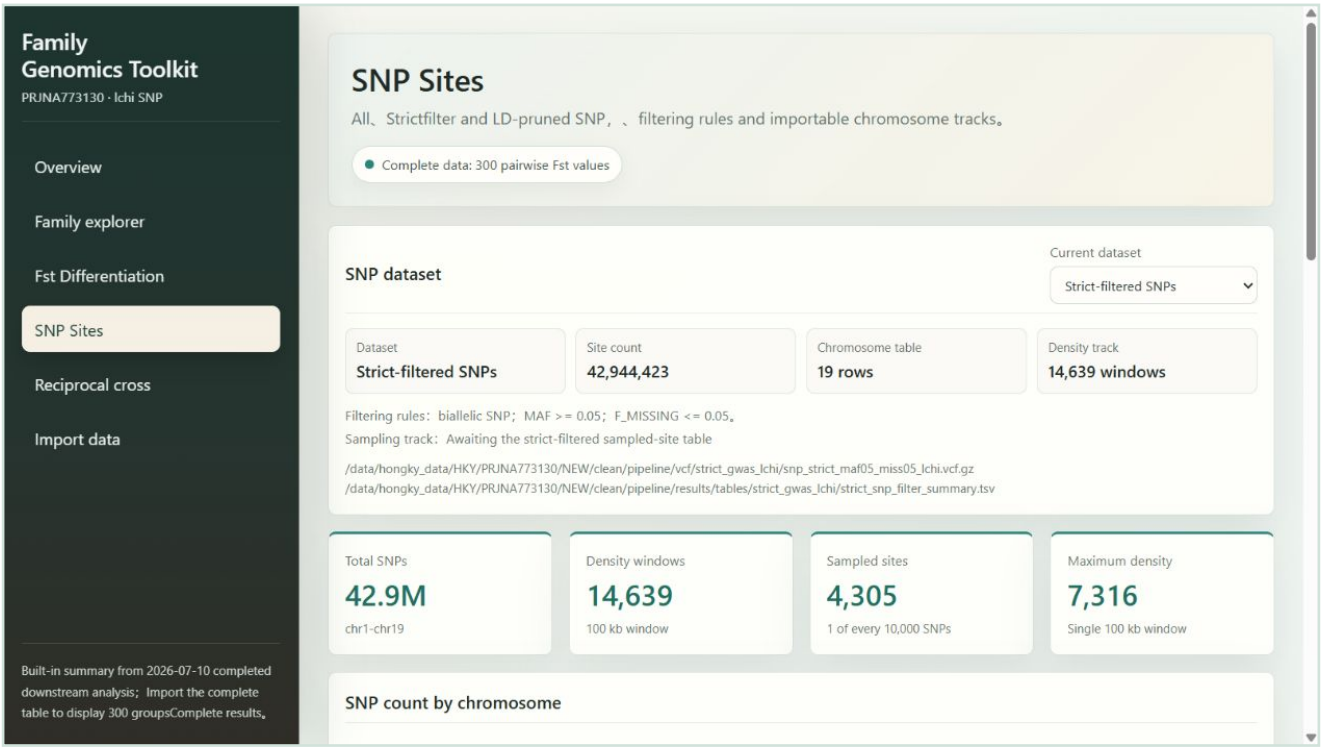
Home Page → Analysis Tools → Genome & Variation → Population SNP Atlas (tool key snp); Open in New Window to open snp-analysis.html#snp. The top level can also Download filter summary.

Open website/independent page

Function

Browse 233 resequencing samples, pre-calculated population statistics of 25 hybrid families, as well as candidate/strictly filtered/LD-pruned SNP counts, densities and sampling sites; local result tables in the same format can be imported.

2026-10-21 has been opened from the online address and taken a screenshot.



Population SNP Atlas · Population variation browsing

real online screenshot: 2026-10-21.

Input and parameters

- Sidebar view: Overview, Family explorer, Fst Differentiation, SNP Sites, Reciprocal cross, Import data (webpage translated from Chinese source copy into English).
- Overview Sort: By Ho, By F value, By within-family kinship, By sample count. Family explorer: Family drop-down; Metric options include Mean Ho, Inbreeding coefficient F, Within-family kinship, and Sample count.
- Fst view: Focus family, Direction (Highest differentiation / Lowest differentiation / Related to focused family), Rows to show (10 / 15 default / 25).
- Current dataset of SNP view: Strict-filtered SNPs (default, strict), LD-pruned SNP (ldpruned), All imputed candidate SNPs (full). Chromosome drop-down; Display selectable All chromosomes / High-density windows; Start Mb slider; Span Mb selectable 5 / 10 default / 25 / 50 / full chromosomes.
- Color of Reciprocal cross by: Family, Maternal species, Reciprocal cross. Choose files of Import data support multiple selection of .tsv/.csv/.txt/.eigenvec/.eigenval, and identify the format by file name.

Sample process

has built-in data viewing path: SNP Sites → Strict-filtered SNPs → chr1 → Span Mb 10; Family explorer optional L_FY1. The import example only refers to the format: strict_maf05_miss05_chr_counts.tsv requires CHROM/N_SNP, snp_density...tsv requires CHROM/START/END/N_SNP, snp_site_sample...tsv requires CHROM/POS/ORDINAL_IN_CHR, and the real file is generated by user analysis.

Step by step operation

1. Enter Tools and click Open in New Window for a more complete widescreen chart; the default anchor enters SNP view.
2. Select Strict-filtered SNPs in the Current dataset, check the filtering rules and total number; switch LD-pruned or All imputed candidate SNPs to compare the data level.
3. Select Chromosome, such as chr1; select Display, high-density window as needed, drag Start Mb and change Span Mb to view the sampling site track.
4. Select a family in Family explorer (for example, preset L_FY1), and select Metric to compare Ho, F, kinship within the family, and the number of samples.
5. Select the Focus family in the Fst view, select Related to focused family in the Direction, and view the corresponding pair statistics; or view the highest/lowest combination.
6. View the PC test table and PCA scatter points in Reciprocal cross, change the Color by check grouping; the significant color is based on the page p<0.05 threshold.
7. When you need to import, go to Import data → Choose files, select the calculated results table listed on the page, view the identification message and Current data volume, and then return to the corresponding view.

How to read the result

- The total number of data realized on the current page: All imputed candidate SNPs=67,104,478; Strict-filtered SNPs=42,944,423, rules biallelic, MAF≥0.05, F_MISSING≤0.05; LD-pruned=8,263,003, using --indep-pairwise 50 5 0.2.
- The density is a 100kb window, the locus track is sampling, not all SNPs are displayed one by one; there is no single gene/single SNP ID search box.
- Overview/Family explorer displays the pre-calculated sample number, Ho (observed heterozygosity), F and kinship; the Fst heat map reflects the pre-calculated family differentiation, not the causal effect.
- Reciprocal cross shows precomputed Wilcoxon test with PCA; PCA sample separation and p<0.05 cannot alone be interpreted as causal evidence of maternal effects.
- Download filter summary downloads the filter statistics TSV; there is no button to rerun filtering or export VCF online on this page.

Frequently Asked Questions and Usage Restrictions

- This is precomputed results browsing and local summary table import, does not accept VCF and recalculates MAF, LD, PCA, Fst or affinity matrices online.
- Import based on file name keyword identification: pairwise_fst_summary, heterozygosity_by_cross, within_family_kinship, family_size_summary, reciprocal_pc_tests, info, eigenvec, chr_counts, snp_density, snp_site_sample, etc.; just changing the extension is not enough for compatibility.
- The absolute path to the server displayed on the interface is used to indicate the data source; to obtain files, you must use the clearly provided public download entrance.
- Page residual copy 66/300, Awaiting import or old sampling frequency may be inconsistent with the embedded data, and the current count/actual chart and selected data set should prevail.
- The imported data is stored in the browser page memory; the current page implementation does not implement server storage or automatic persistence and should be re-imported after refreshing.

Ortholog Analysis · Ortholog lookup

genome and variation | Preset homology record query

Entrance

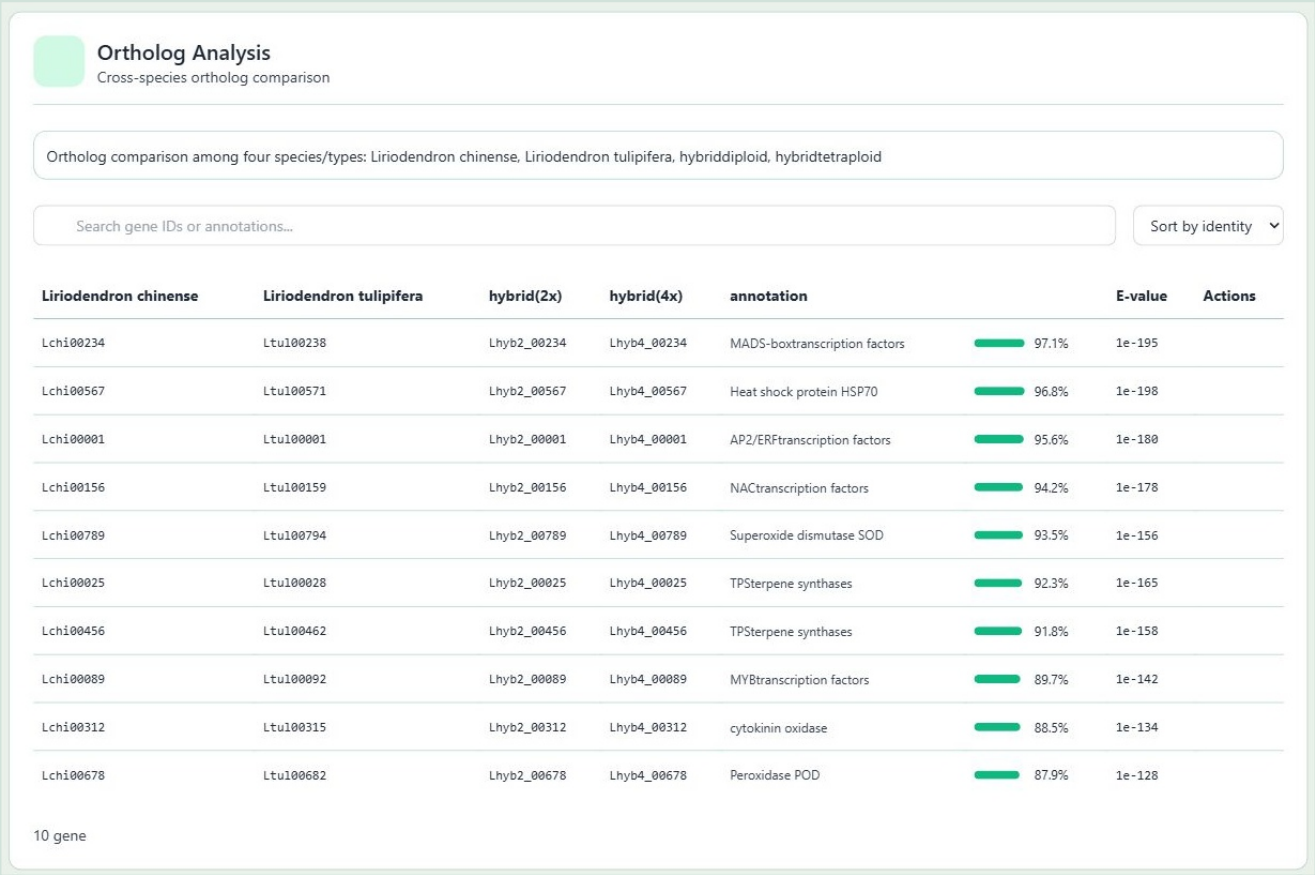
Home Page → Analysis Tools → Genome & Variation → Ortholog Analysis (tool key ortholog)

Open website/independent page

Function

View the homologous corresponding records of Chinese Liriodendron, North American Liriodendron and hybrid diploid/tetraploid preset on the page.

2026-10-21 has been opened from the online address and taken a screenshot.



Ortholog Analysis · Same origin corresponding query

real online screenshot: 2026-10-21.

Input and parameters

- Search gene IDs or annotations...: Any column of gene IDs or annotation keywords can be filtered immediately after input without clicking the search button.
- Sort drop-down: Sort by identity, E-valuesort, Gene IDsort (actual copywriting is implemented on the current page).

Sample process

Enter the preset ID Lchi00001 to locate the corresponding rows of Ltul00001, Lhyb2_00001, and Lhyb4_00001 in the current page; this is only an example of existing data navigation.

Step by step operation

1. In Search gene IDs or annotations..., enter Lchi00001, or TPS and other annotation keywords.
2. Select identity descending order, E-value ascending order or Gene ID sorting as needed.

3. Horizontally browse the gene correspondence, annotation, identity percentage and E-value of the four types of materials.
4. Click the eye icon in the Actions column to expand the corresponding details; use the X icon to close.

How to read the result

- Each row is the preset correspondence between four types of materials; details display functional annotation, sequence percentage, alignment length (aa) and E-value.
- The bottom of the page displays the number of filtered gene rows; similarity and E-value are ready-made fields in the record, and there is no comparison in this run.

Frequently Asked Questions and Usage Restrictions

- Currently only 10 preset homologous corresponding records are provided; there is no sequence upload, online orthogonal inference or two-way best hit calculation.
- No export button; misses cannot be interpreted as genes not homologous across species.
- These records are located in the implementation example data area of the current page. The real data source/version needs to be confirmed separately; similar sequences cannot directly determine the orthogonal relationship based on identity alone.

Synteny Analysis · Syntenic block browser

genome and variation | Preset collinear block browsing

Entrance

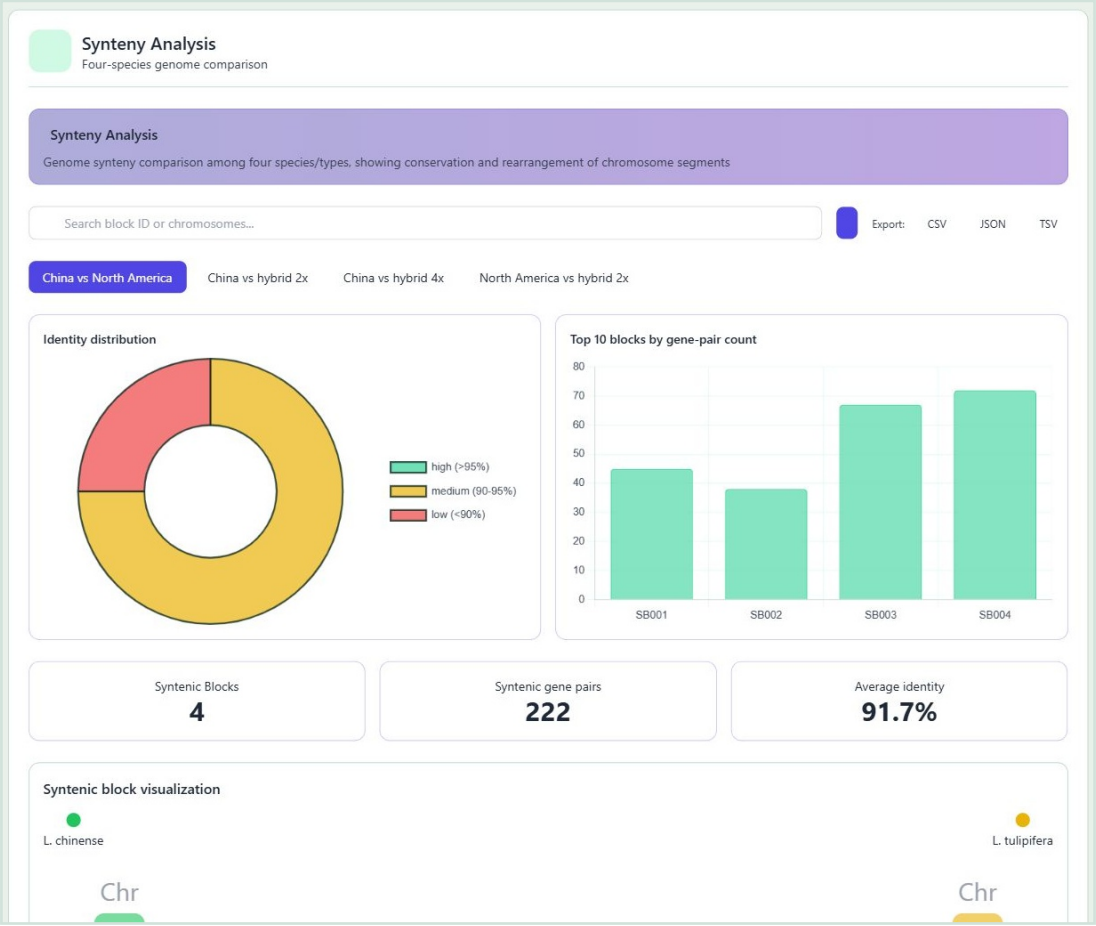
Home Page → Analysis Tools → Genome & Variation → Synteny Analysis (tool key synteny). Not to be confused with the separate resource page for Omics Data → Genomics → Comparative Genomics.

Open website/independent page

Function

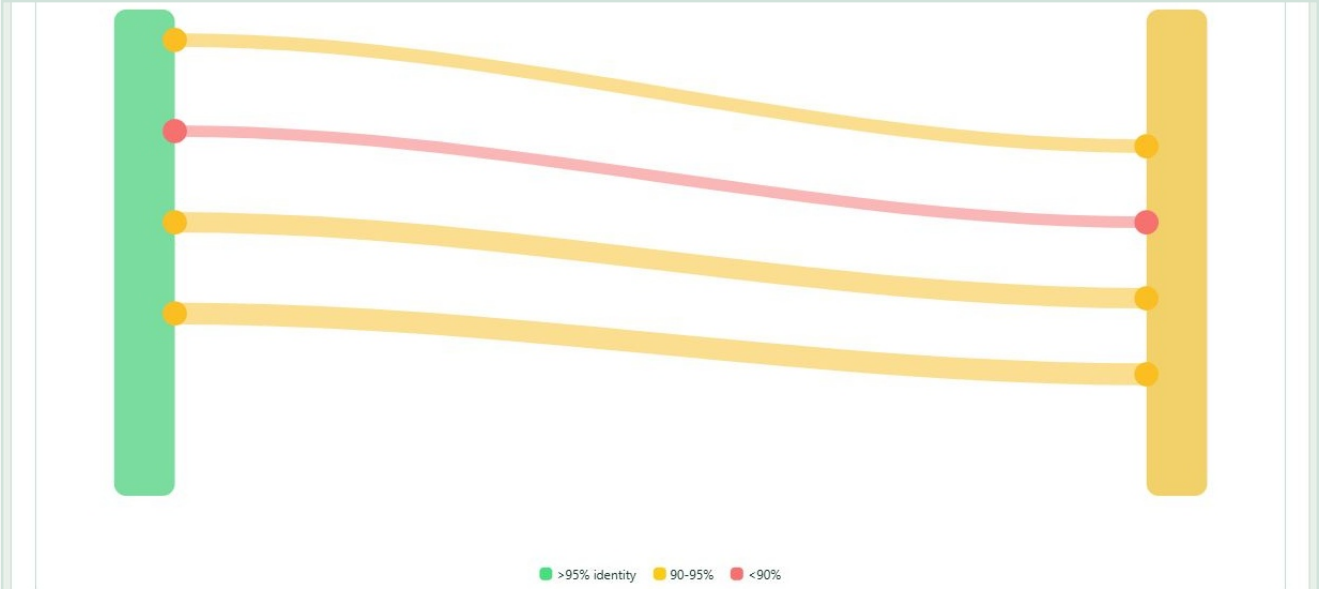
presets collinear blocks on the species pair and chromosome filtering page, views the block table, statistics and graphical connections, and exports the block list.

2026-10-21 has been opened from the online address and taken a screenshot.



Synteny Analysis · Collinear block query

real online screenshot: 2026-10-21.



Block ID	L. chinense	L. tulipifera	Gene pairs	identity
SB001	Chr01:1.0-5.0Mb	Chr01:1.2-5.3Mb	45	92.3%
SB002	Chr01:8.0-12.0Mb	Chr02:2.0-6.0Mb	38	89.7%
SB003	Chr02:1.0-8.0Mb	Chr02:8.0-15.0Mb	67	91.2%
SB004	Chr03:2.0-10.0Mb	Chr03:1.5-9.5Mb	72	93.5%

Synteny Analysis · Collinear block query

real online screenshot: 2026-10-21.

Input and parameters

- Species pair buttons: China vs North America, China vs hybrid 2x, China vs hybrid 4x, North America vs hybrid 2x.
- Search block ID or chromosomes...: Enter the block ID or chromosome name to filter in real time.
- The chart switch is the pie-chart icon button; the export buttons are CSV, JSON, and TSV.

Sample process

China vs North America → Enter SB001 → Click on the table row → CSV/TSV. SB001 on the current page is the block corresponding to Chr01. The example does not mean re-completing the whole-genome collinearity inference in real time.

Step by step operation

1. Click on the species pair first, such as China vs North America.
2. Enter SB001 or Chr01 in Search block ID or chromosomes... and read the filtered Syntenic Blocks, Syntenic gene pairs and Average identity.
3. View the Identity distribution pie chart and block gene-pair bar chart; you can use the chart icon to hide or show it.
4. Click on the connection line in Syntenic block visualization or the block table row below to view the starting and ending Mb positions, gene pairs and identity.
5. Use CSV, JSON or TSV to export the currently filtered block list and save species pairs, chromosome intervals and filter conditions.

How to read the result

- The block table displays Block ID, chromosomes at both ends and Mb intervals, Gene pairs, and identity. The statistics card recalculates the quantity, geneCount sum and identity simple mean based on the current filtered block.
- Identity distribution is divided into three levels: >95%, (90,95], and ≤90; the <90 text in the legend is slightly inconsistent with the boundary of the code ≤90.
- The exported fields include block_id, species1/2, chr1/2, start/end_mb, gene_pairs, and identity_percent.

Frequently Asked Questions and Usage Restrictions

- Currently browsing 7 preset collinear blocks; there is no entry to upload GFF/FASTA, execute MCScanX or recalculate collinearity. The connection positions of
- Syntenic block visualization are arranged according to the block index and are not drawn according to the proportion of real chromosome coordinates; it is a schematic diagram, and the scientific research coordinates should be seen in the table.
- Top 10 blocks by gene-pair count actually directly takes the top 10 items of the current list without reordering by geneCount. The manual does not guarantee that this is the sorted Top10.
- There are only 4 listed combinations of species pairs, it cannot be implied that all 6 pairwise combinations are open.

Gene Age Query · Gene age records

genome and variation | Preset age level record query

Entrance

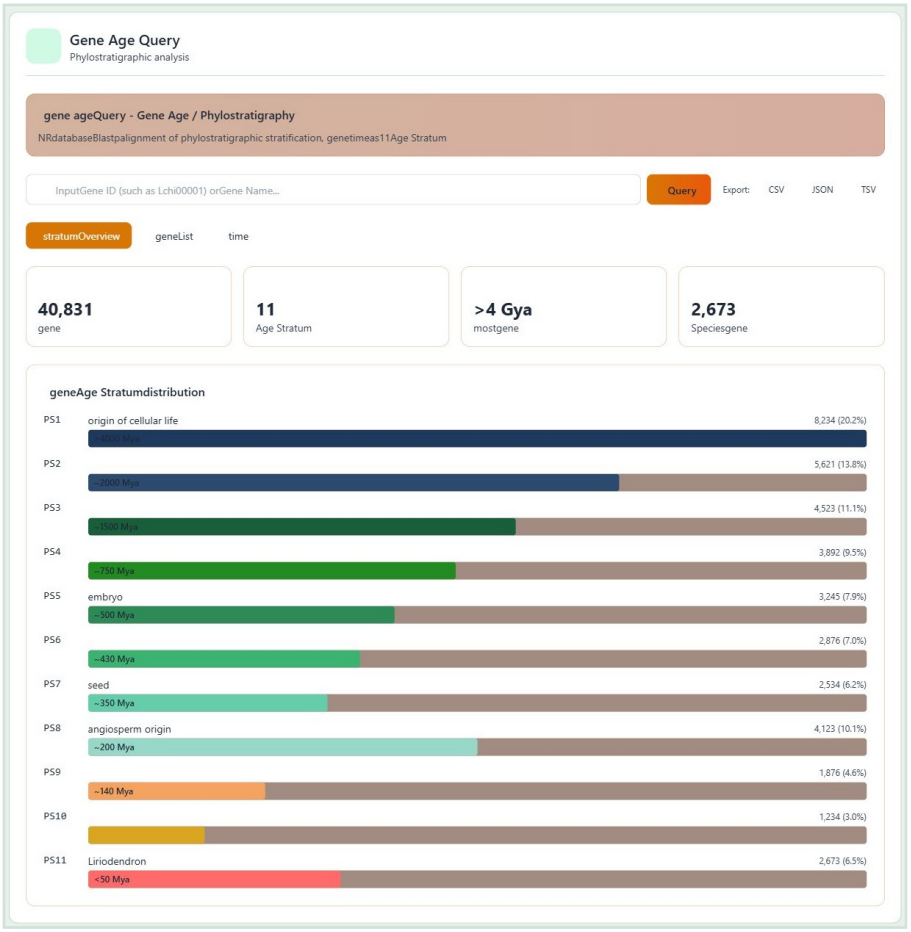
Home Page → Analysis Tools → Genome & Variation → Gene Age Query (tool key geneage)

Open website/independent page

Function

Query the preset gene age level records on the page by gene ID, name or annotation, and browse the overview and time graph of 11 phylostrata.

2026-10-21 has been opened from the online address and taken a screenshot.



Gene Age Query · Gene Age Record

real online screenshot: 2026-10-21.

Input and parameters

- InputGene ID (such as Lchi00001) orGene Name...: Enter a single ID, gene name or annotation fragment.
- Query button or Enter to execute query; no batch ID list parsing.
- The actual copy of the view button: stratumOverview, geneList, time. Export CSV, JSON, TSV.

Sample process

Lchi00001 → Query → geneList → Click on the target row. You can also enter preset annotation words such as TPS or MYB; the multi-ID comma list cannot be regarded as a supported batch input.

Step by step operation

1. Enter the known page example Lchi00001, then click Query or press Enter.

2. Click geneList to switch to the list and check whether Found N of gene / Searchresults N and target ID appear. You cannot just check whether the total list has content.
3. Click on the gene row to expand functional annotation, Age Stratum, Taxon and best hit details.
4. Click stratumOverview to view the 11-layer statistics; click a layer to view the level description and preset function words; click time to view the instructions from ancient times to the present.
5. Use CSV/JSON/TSV to export the current hit query; use clear results when you need to query again, and then enter the next ID.

How to read the result

- List fields: Gene ID, Gene Name, Age Stratum, Taxon, mostalignment, E-value, Identity; details are supplemented by preset age and best hit species.
- Lchi00001 in the current page corresponds to PS8/Magnoliopsida, which is an example of checking existing records; the page will not run BLAST or phylostratigraphy for the input sequence.
- The level statistics are preset global numbers; sampleGenes has only 24 entries, and the two are not in the same coverage.

Frequently Asked Questions and Usage Restrictions

- Currently searching in 24 preset gene records; the total number and function words of 11 levels are existing statistics, not the whole genome results calculated in real time after input.
- When there is no hit in the query, searchResults is empty, and the list and export logic will fall back to display/export all sampleGenes without a clear 0 result warning; therefore, do not regard the entire table as a hit when there is no Found prompt.
- Clicking Query will not automatically switch geneList. You need to manually click geneList to view the query list.
- Gene age is a preset homology search level and rough age description, not the precise origin time of a single gene; the original basis for homology search coverage, thresholds and classification levels must be verified.

Expression Atlas · Gene expression query

expression and regulation | Preprocessing expression matrix query

Entrance

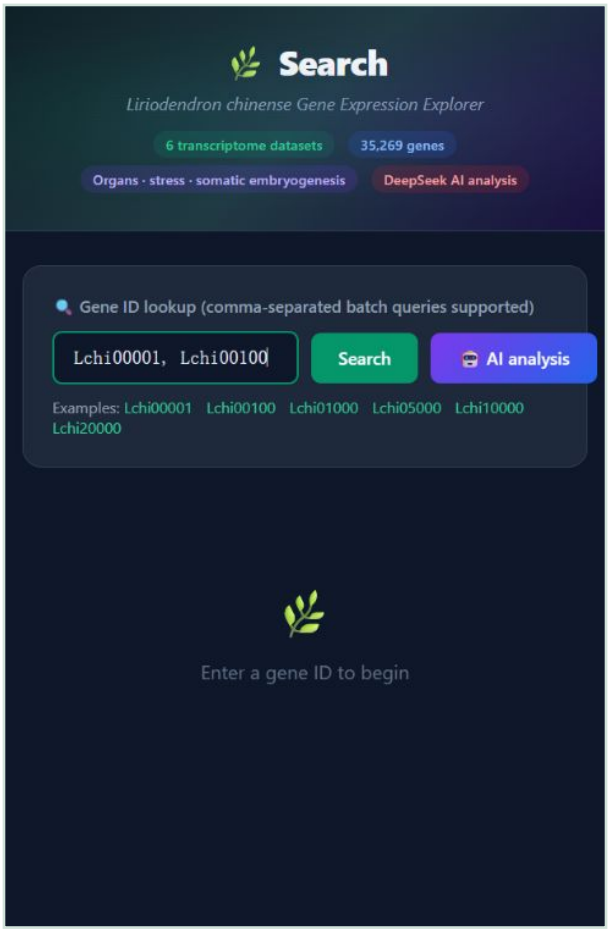
Analysis Tools → Expression & Regulation → Expression Atlas; you can also click Open Expression Atlas from the transcriptome directory, or directly open gene_expression_explorer.html. Open in New Window on the upper right side of the embedded page can be viewed independently.

Open website/independent page

Function

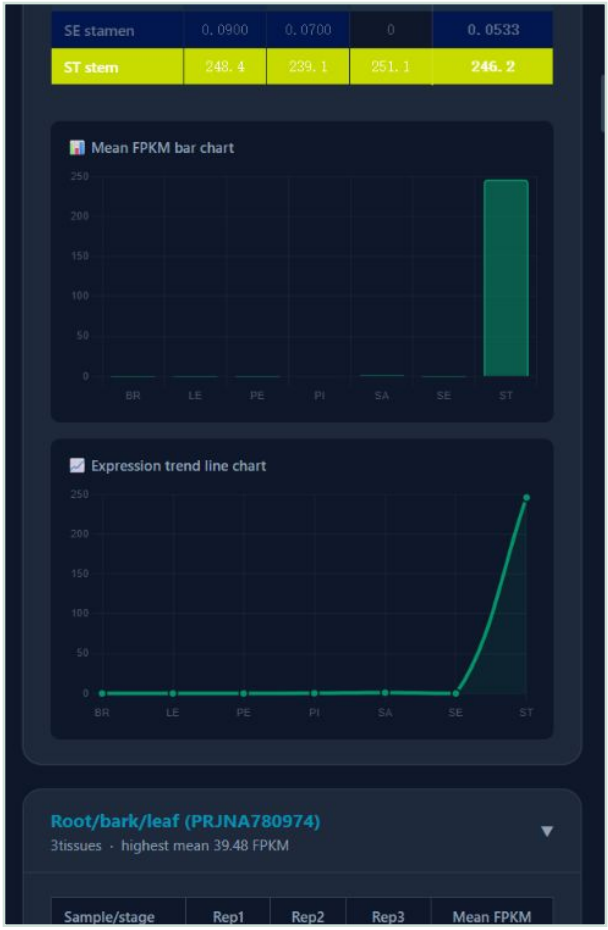
queries the FPKM expression of six sets of preprocessed transcriptomes by Lchi gene number, and checks the repetitions, grouping means and expression trends.

2026-10-20 Lchi00001 and Lchi00100 have been actually queried, and screenshots of expression tables, histograms and trend charts have been obtained. This section reuses the screenshot; no AI request was submitted in this round.



Expression Atlas · Gene Expression Query

real online screenshot: 2026-10-20.



Expression Atlas · Gene Expression Query

real online screenshot: 2026-10-20.

Input and parameters

- Complete, case-correct gene ID, such as Lchi00001; supports English commas, Chinese commas or blanks to separate multiple genes.
- Only the built-in expression matrix is queried here; FASTQ is not uploaded, and gene names or protein sequences are not received.

Sample process

Input Lchi00001,Lchi00100 → Search → View the organ expression and stress time series of each gene. If you only want to analyze Lchi00001, change to query only Lchi00001 and click AI analysis.

Step by step operation

1. Enter Expression Atlas and wait for the Loading the gene expression database mask to disappear.
2. Fill in Lchi00001 in the Gene ID lookup input box, click Search or press Enter; you can also click on any of the example IDs below to query immediately.
3. Batch query example: Lchi00001, Lchi00100, Lchi01000. Data sets with data are listed under each hit gene.
4. Check the corresponding Sample/stage, Rep1, Rep2, Rep3 and Mean FPKM in the six categories of organs Different, Root/bark/leaf, Somatic embryogenesis, Cold/heat stress, Drought stress, Cold-stressed seedlings.
5. Click the dataset title to expand or collapse. Move your mouse over the bars of the Mean FPKM bar chart or the data points of the Expression trend line chart to see the complete group names and values.
6. When AI interpretation is required, it is recommended to query a gene separately and then click AI analysis. The page first requests the configured public AI agent; if the service is unavailable and the machine does not have a key, a pop-up window will appear asking for the DeepSeek API Key. Fill in your own valid key and try again; you can view expression results without using AI.

7. If Gene not found is displayed, check the spelling and case of the ID, and then use the page example Lchi00001 to confirm the query format.

How to read the result

- Six sets of data: PRJNA559687 (7 organs), PRJNA780974 (root/bark/leaf), PRJNA1177383 (11 individual embryogenesis stages), PRJNA679089 (cold/heat stress), PRJNA679101 (drought), PRJNA761222 (seedling cold stress).
- Displays three replicate values, mean FPKM, colored table, bar chart, and line chart by data set. Coloring is a representation of relative expression within the data set.
- Displays textual interpretation in the gene's DeepSeek AI Analysis panel when the AI succeeds; displays an error when the API call fails, and falls back to Rule-based quick analysis (summary of rules by data set highest expression grouping).

Frequently Asked Questions and Usage Restrictions

- FPKM expression display does not equal differential expression testing; the page does not have DESeq2/edgeR calculations, significance tests, P values, or FDRs. Direct comparison of absolute FPKM values across studies requires a combination of experimental conditions.
- When querying in batches, AI analysis actually analyzes the last successfully hit gene instead of all query genes; the manual should prompt to run AI on a single gene.
- The page says Auto-connected · no configuration required, but the actual situation still depends on the public proxy or valid native DeepSeek API Key, network and browser cross-domain conditions, and it cannot be claimed that AI is guaranteed to be available.

-The existing page does not have a CSV or image export button to express the results; if the original matrix is required, the actual provided file should be obtained from the data download portal.

- The page requires the browser to support DecompressionStream; if loading fails, you can refresh or change to a newer browser.

miRNA Target · miRNA target catalog

expression and regulation | Preset miRNA and target directory

Entrance

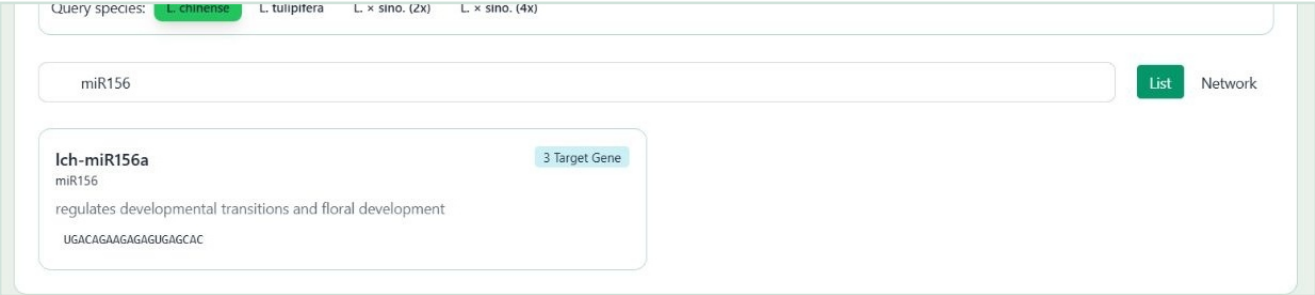
Analysis Tools → Expression & Regulation → miRNA Target□

Open website/independent page

Function

Browse the built-in miRNA sequences, families, target annotations, existing target gene IDs and tissue expression records of four materials.

2026-10-21 has been opened from the online address and taken a screenshot.



miRNA Target·miRNA target directory

real online screenshot: 2026-10-21.

Input and parameters

- First select Liriodendron tulipifera, Liriodendron tulipifera, hybrid diploid or hybrid tetraploid in Query species.
- The search box supports miRNA ID, family, functional description or annotated name of the target, e.g. lch-miR156a, miR156, SPL2. Matching is not case sensitive.

Sample process

Query species Select Liriodendron chinense → Search for miR156 → Click on lch-miR156a → View Lchi00234, Lchi00312, Lchi00456 and SPL2/SPL6/SPL9 entries.

Step by step operation

1. Enter the miRNA Target and select the material in Query species.
2. Enter miR156 in Search miRNA ID, family, or target gene...; the list will be filtered instantly as you enter, no submit button is required.
3. Click the miRNA card in List mode to open the details of the entry.
4. Read sequence, family, function, Predicted target gene IDs, Target gene annotation and Tissue expression (TPM) in details.
5. Click Network to switch the relationship diagram. Click the miRNA dot in the figure to open the same details; click List again to return to the list.
6. Modify or clear the search terms to restore more entries; click × in the upper right corner of the details to close the details.

How to read the result

-miRNA card shows number, family, sequence, functional description and number of existing targets.

- Details show the predicted target gene number and target name, and list the expression in leaf, stem, root, and flower tissues.
- Network connects target annotation names with miRNA nodes to assist in browsing existing relationships in the directory.

Frequently Asked Questions and Usage Restrictions

- The current interface browses preset records, and there is no button or back-end calculation for re-prediction after entering new miRNA/transcript sequences.
- The search implementation did not retrieve the Predicted target gene IDs array; entering numerical IDs such as Lchi00234 may not hit, you should first search with target names such as miRNA ID, family or SPL2, and then check the number in details.
- Network displays at most the top 6 miRNAs in the screening results, and each can draw up to 3 target names, which cannot be regarded as a complete network.
- There are only 10 built-in entries for Liriodendron tulipifera and 4 for each of the other three materials; the page has no data source, prediction score, experimental verification evidence or export button and should be browsed as an existing directory.
- Some records have different numbers of targets and targetGenes, and the one-to-one correspondence between each ID and target name cannot be inferred based on the display order alone.

GO/KEGG Annotation · Functional annotation lookup

expression and regulation | GO/KEGG annotation query available

Entrance

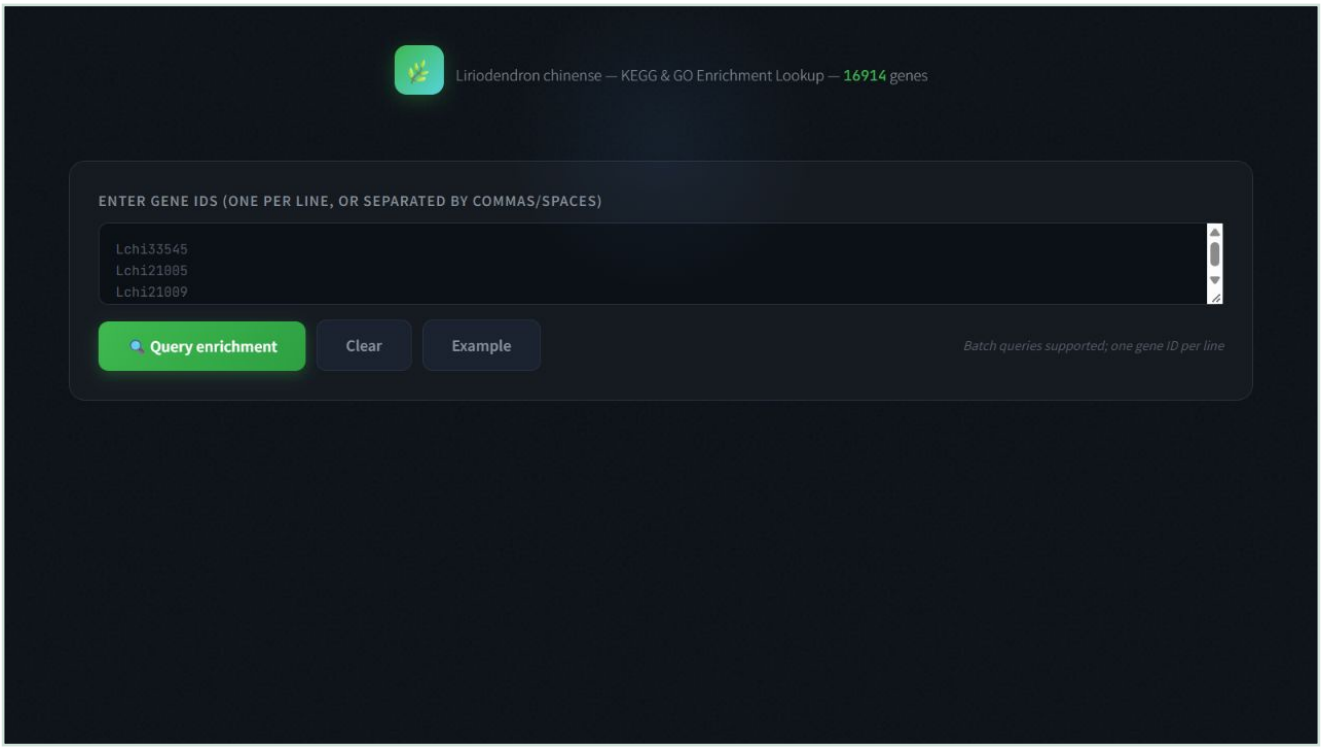
Analysis Tools → Expression & Regulation → GO/KEGG Enrichment; corresponding to the independent page tool.html.

Open website/independent page

Function

Batch query existing GO and KEGG annotations, jump terms and pathway details according to Liriodendron tulipifera Lchi gene ID. The page name is reserved as Enrichment, and the actual function is annotation query.

2026-10-21 A screenshot of the online input page has been obtained. The browser response timed out after clicking Example, and no screenshot of the query results was obtained in this round; the following result fields are explained based on the current page implementation.



GO/KEGG Enrichment · Function annotation query

real online screenshot: 2026-10-21.

Input and parameters

- Complete, case-correct Lchi gene numbers, one per line, or separated by commas, spaces, tabs, or semicolons.
- e.g. Lchi33545, Lchi21005, Lchi21009; do not enter a GO number or pathway number in place of the gene ID.

Sample process

Click Example → Automatically fill in Lchi33545, Lchi21005, Lchi21009, Lchi21047, Lchi16252 → Expand Lchi33545 → Click a GO number or KEGG pathway number.

Step by step operation

1. Enter GO/KEGG Enrichment. Paste the list of gene IDs into the Enter gene IDs (one per line, or separated by commas/spaces) text box.
2. Click Query enrichment, or press Ctrl+Enter. When using it for the first time, you can directly click Example to automatically load and query 5 example genes.

3. View the top Genes queried, With GO annotations, With KEGG annotations statistics, distinguish the number of input entries and the number of entries with corresponding annotations.
4. Click the title of a gene card to expand or collapse. When querying no more than 5 items, the hit card will be expanded by default; if there are more than 5 items, you need to click to expand.
5. In GO annotations, click the number starting with GO: and go to AmiGO to view the definition of the term; in KEGG annotations, click the K number to view the KO entry, and click the ko/map pathway number to open the KEGG pathway or pathway map.
6. Click Clear to clear the input, results and statistics.

How to read the result

- List of GO annotation numbers for each hit gene, along with KEGG KO numbers, functional names, pathways, and pathway map external links.
- Missed entries display No annotation information found for this gene; when a certain type of annotation is missing, no GO or no KEGG information is displayed.

Frequently Asked Questions and Usage Restrictions

- The gene-by-gene reads on this page have GO and KEGG annotations; there are no background gene sets, hypergeometric or Fisher tests, and no P-values, corrected P-values, FDRs, fold enrichment or enrichment bubble plots. The results on this page cannot be used as statistical enrichment conclusions.
- Only the Liriodendron tulipifera ID in the built-in annotation library can be hit; no annotation found does not mean that the gene has no biological function.
- Entering duplicate IDs will not automatically remove duplicates, but will be counted and displayed repeatedly; it is recommended to delete duplicates before pasting.
- There is no result export button on the existing page; GO and KEGG details depend on accessing the corresponding external website.

Co-expression Network · Co-expression network browser

expression and regulation | Preset co-expression web browsing

Entrance

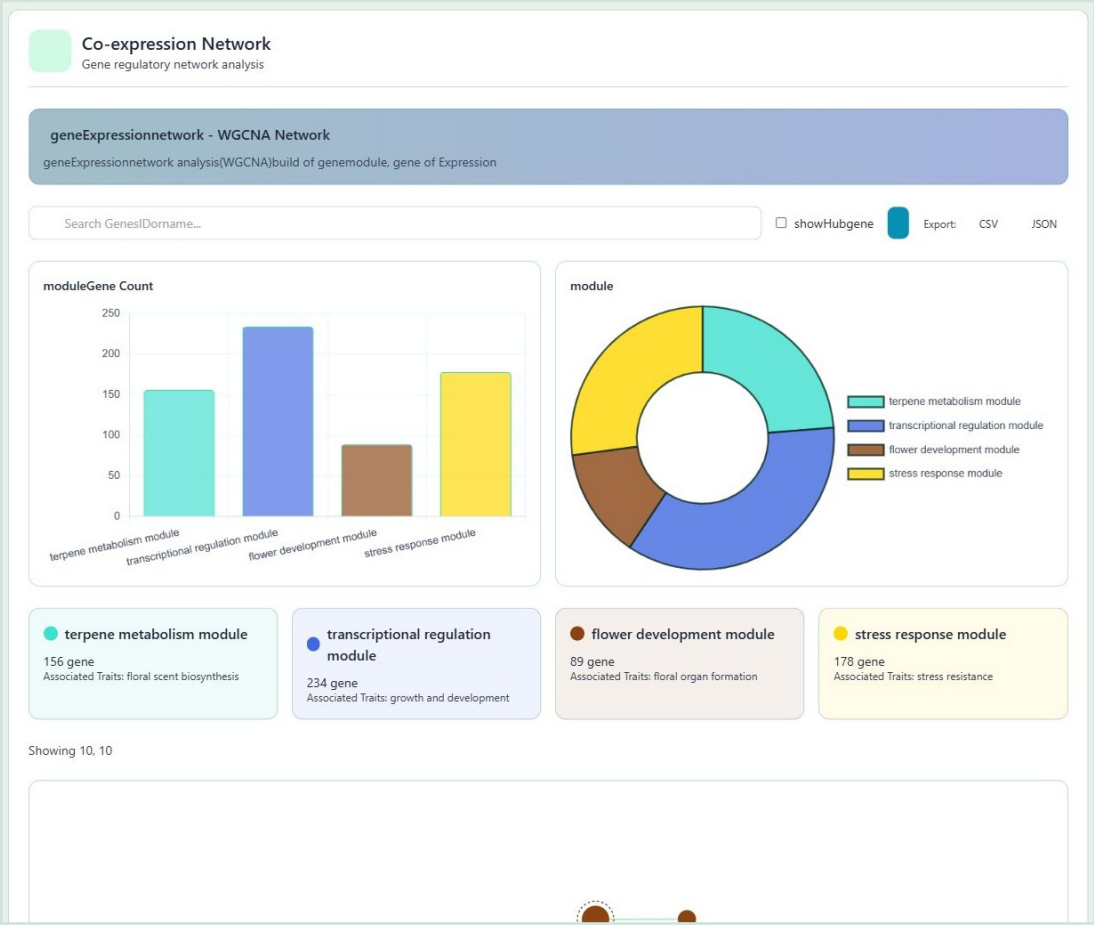
Analysis Tools → Expression & Regulation → Co-expression Network

Open website/independent page

Function

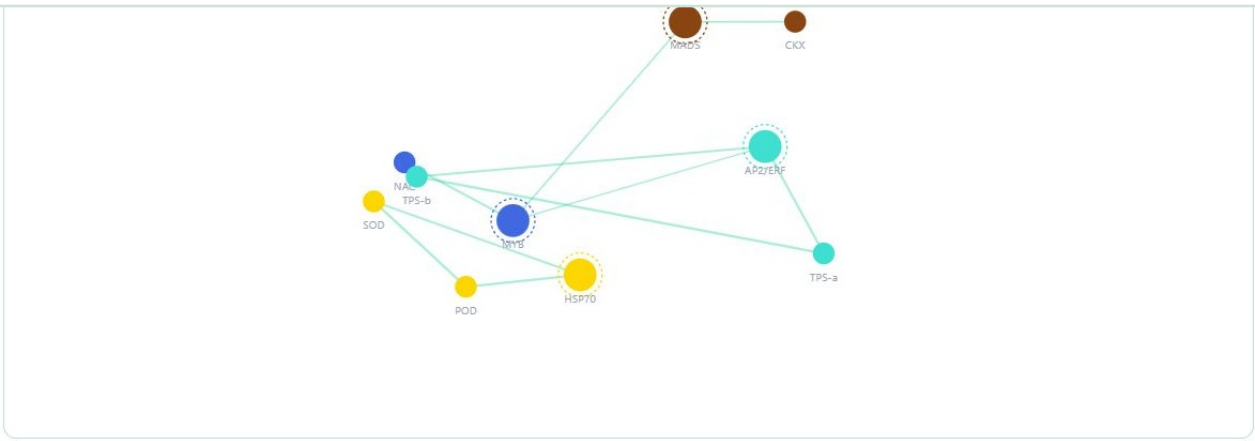
Browse the built-in Liriodendron tulipifera co-expression network, modules, Hub genes, module member correlation kME and associated traits, and export the filtered node table.

2026-10-21 has been opened from the online address and taken a screenshot.



Co-expression Network · Co-expression Network

real online screenshot: 2026-10-21.



Gene ID	Gene Name	module	Hubgene	kME
Lchi00001	AP2/ERF	terpene metabolism module		0.92
Lchi00025	TPS-a	terpene metabolism module	-	0.78
Lchi00089	MYB	transcriptional regulation module		0.89
Lchi00156	NAC	transcriptional regulation module	-	0.72
Lchi00234	MADS	flower development module		0.95
Lchi00312	CKX	flower development module	-	0.68
Lchi00456	TPS-b	terpene metabolism module	-	0.81
Lchi00567	HSP70	stress response module		0.88
Lchi00678	POD	stress response module	-	0.75
Lchi00789	SOD	stress response module	-	0.71

Co-expression Network · Co-expression Network

real online screenshot: 2026-10-21.

Input and parameters

- Gene ID or name keyword, e.g. Lchi00001, AP2/ERF, TPS.
- You can click on the module card to filter, or you can check showHubgene to only retain Hub genes. No need to upload files.

Sample process

Click terpene metabolism module → Search TPS → View TPS-a/TPS-b nodes. After clearing the search, check showHubgene to view the AP2/ERF Hub node of the module, and then click CSV export.

Step by step operation

1. Enter the Co-expression Network and first observe the four module cards and the default network.
2. Enter TPS in Search GenesIDorname..., the list and node graph will be filtered immediately; clear the search box to restore.
3. Click the terpene metabolism module, transcriptional regulation module, flower development module, or stress response module card to view only that module. Click the same card again to cancel module filtering.
4. Check showHubgene to retain only Hub nodes. Dashed circles and tabular asterisks outside nodes are used to designate hubs.
5. Click the network node or table row to view the Gene ID, module, kME, Associated Traits and connected genes in detail below.
6. Click the bar chart icon on the toolbar to hide or display the module quantity chart and module proportion chart.
7. Click CSV or JSON to save the current filtered node information as coexpression_nodes.csv or coexpression_nodes.json.

How to read the result

- Module number histogram, module proportion chart, four module cards, network diagram and node table.
- Node table fields include Gene ID, Gene Name, module, Hubgene, and kME; CSV/JSON export also includes module associated traits.
- Node details lists the directly connected genes of this node in the built-in complete network.

Frequently Asked Questions and Usage Restrictions

- The currently provided built-in network only has 10 nodes and 10 edges; the number of genes (156, 234, 89, 178) of the four module cards is not equal to the number of nodes that can be browsed or exported one by one on the page.
- There is no entry point for uploading expression matrices, selecting soft thresholds, re-executing WGCNA or recalculating modules. The network layout is a front-end display, and the node positions may change when refreshing or filtering.
- The page edge count is only updated with module filtering and is not reduced simultaneously with keyword/Hub filtering. It may be higher than the current actual number of drawn edges; do not regard this number as the number of visible edges after filtering.
- CSV/JSON button only exports nodes. Although the current page implementation prepares edge data, the page does not have an edge table download button.
- The current page implementation does not provide the original expression matrix, network construction parameters and statistical evidence of the built-in network; kME, Hub or trait tags are not equivalent to experimentally verified regulatory relationships.

Heterosis Analysis · Heterosis records

breeding and ploidy | Percentage comparison of preset expression values

Entrance

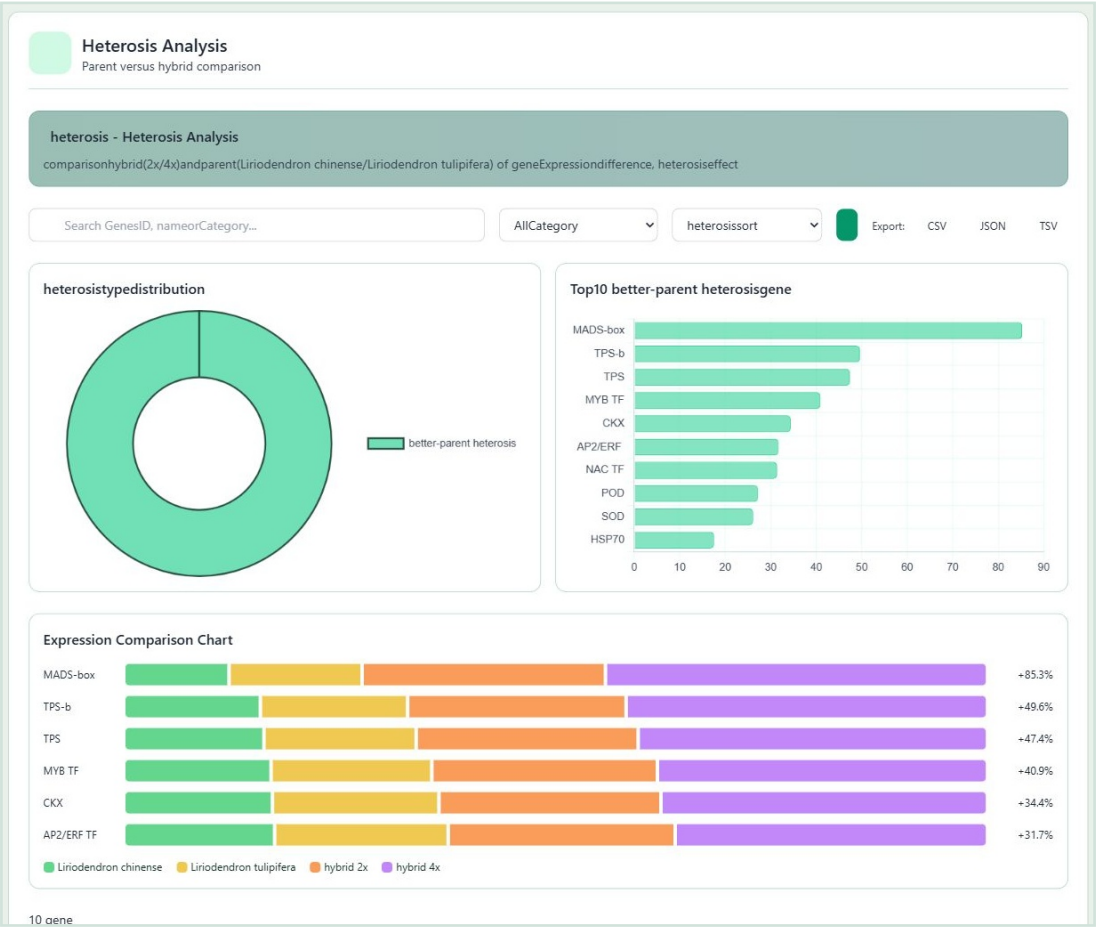
Analysis Tools → Breeding & Ploidy → Heterosis Analysis

Open website/independent page

Function

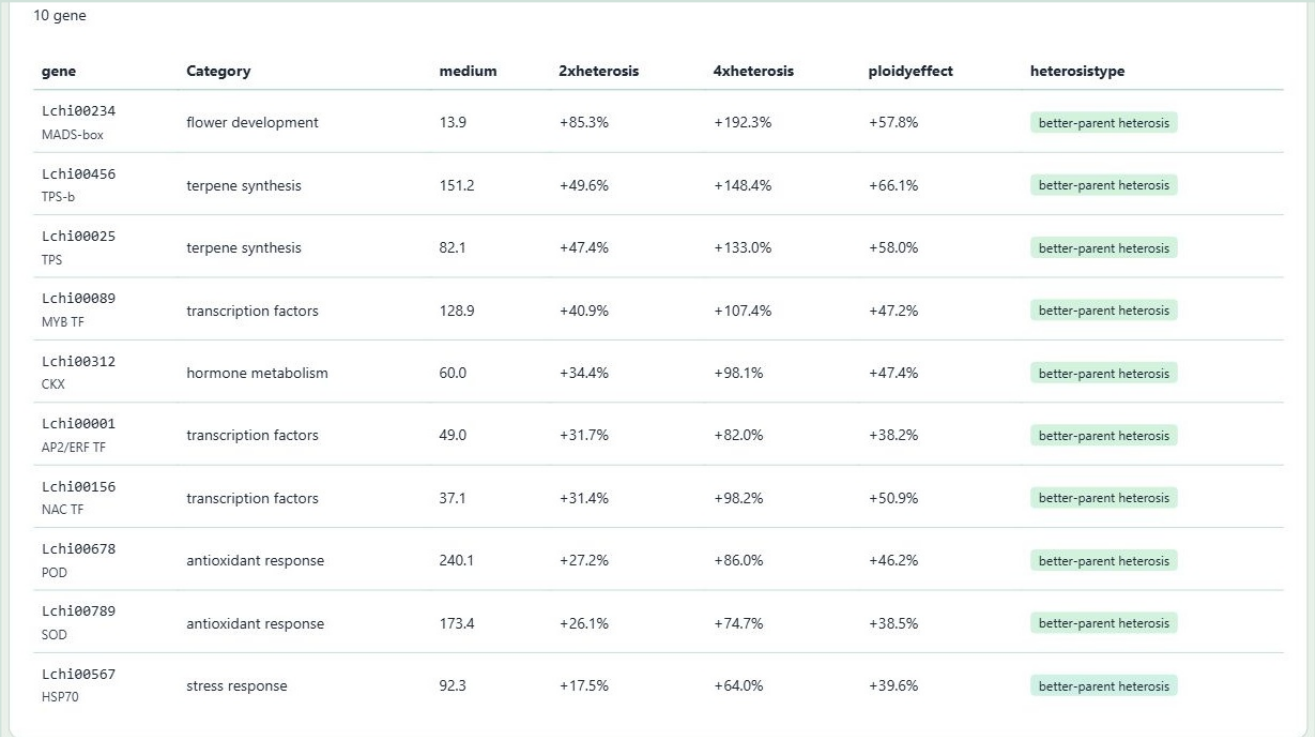
queries the expression values of parents, diploid hybrids and tetraploid hybrids included in the page, and views the mid-parent dominance, super-parent dominance and ploidy effects calculated from these values.

2026-10-21 has been opened from the online address and taken a screenshot.



Heterosis Analysis · Hybrid vigor record

real online screenshot: 2026-10-21.



10 gene

gene	Category	medium	2xheterosis	4xheterosis	ploidyeffect	heterosistype
Lchi00234 MADS-box	flower development	13.9	+85.3%	+192.3%	+57.8%	better-parent heterosis
Lchi00456 TPS-b	terpene synthesis	151.2	+49.6%	+148.4%	+66.1%	better-parent heterosis
Lchi00025 TPS	terpene synthesis	82.1	+47.4%	+133.0%	+58.0%	better-parent heterosis
Lchi00089 MYB TF	transcription factors	128.9	+40.9%	+107.4%	+47.2%	better-parent heterosis
Lchi00312 CKX	hormone metabolism	60.0	+34.4%	+98.1%	+47.4%	better-parent heterosis
Lchi00001 AP2/ERF TF	transcription factors	49.0	+31.7%	+82.0%	+38.2%	better-parent heterosis
Lchi00156 NAC TF	transcription factors	37.1	+31.4%	+98.2%	+50.9%	better-parent heterosis
Lchi00678 POD	antioxidant response	240.1	+27.2%	+86.0%	+46.2%	better-parent heterosis
Lchi00789 SOD	antioxidant response	173.4	+26.1%	+74.7%	+38.5%	better-parent heterosis
Lchi00567 HSP70	stress response	92.3	+17.5%	+64.0%	+39.6%	better-parent heterosis

Heterosis Analysis · Hybrid vigor record

real online screenshot: 2026-10-21.

Input and parameters

- Search GenesID, nameorCategory...: Partial keywords of gene ID, name or category; not case sensitive.
- AllCategory: Select an existing functional category.
- Sort drop-down box: heterosissort, sort by ploidy effect, Gene Namesort.

Sample process

Input Lchi00025 → Click on the corresponding data row of TPS → Compare expression and HPH of parent and Hybrid 2x/4x → Download CSV.

Step by step operation

1. After entering the tool, check all records first. There are currently 10 genes built-in, and there is no need to submit calculation tasks.
2. Enter Lchi00025 or TPS in the search box, and the results will be filtered immediately as you enter; clear the input recovery list.
3. If necessary, use AllCategory to limit categories, and use the sort drop-down box to sort by diploid superparent advantage, ploidy effect, or name.
4. Click the chart icon to show or hide the category pie chart and the superparent advantage bar chart for the top 10 records.
5. Click a record in the Expression Comparison Chart or a row in the data table below to expand the selected gene details.
6. Use the toolbar CSV, JSON or TSV to download the currently filtered and sorted records.

How to read the result

- In the details, L. chinense, L. tulipifera, Hybrid 2x, and Hybrid 4x are the expression values of the two parents and the two hybrid ploidy groups respectively.
- MP=(sum of parental expression)/2; HP=maximum expression of both parents. MPH=(Hybrid-MP)/MP×100%; HPH=(Hybrid-HP)/HP×100%.

- Table 2xheterosis and 4xheterosis columns actually show HPH; ploidyeffect=(4x-2x)/2x×100%. The download contains both MPH and HPH.
- heterosistype is a label that automatically groups pages according to 2x HPH: HPH>0 is better-parent heterosis, -10<HPH≤0 is mid-parent heterosis, and the rest are negative heterosis.

Frequently Asked Questions and Usage Restrictions

- Use the small data built into the current page; there is no entry for users to upload expression matrices or specify experimental groups.

-The result does not provide duplicate samples, statistical tests, or confidence intervals. Percentages represent expression comparisons and cannot alone demonstrate hybrid vigor at the trait level.

- Bar chart title Top10 does not guarantee sorting by superparent dominance: the top 10 items in the current list are used in the figure; when sorting by name or ploidy effect is used, it should be understood according to the current sorting.
- There is insufficient data source description and experimental traceability information. The original data needs to be checked before formal research.

Ploidy Dosage · Ploidy dosage records

breeding and ploidy | Preset ploidy record and ratio display

Entrance

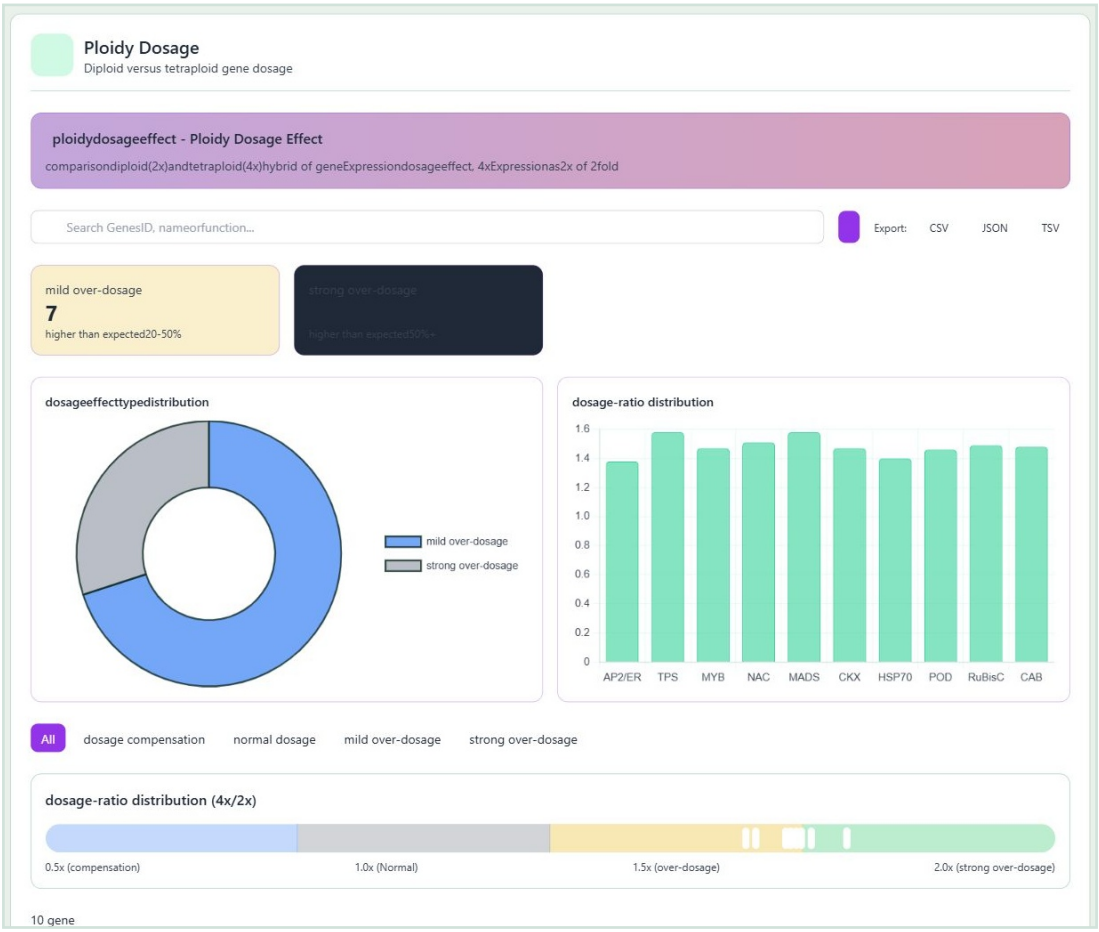
Analysis Tools → Breeding & Ploidy → Ploidy Dosage

Open website/independent page

Function

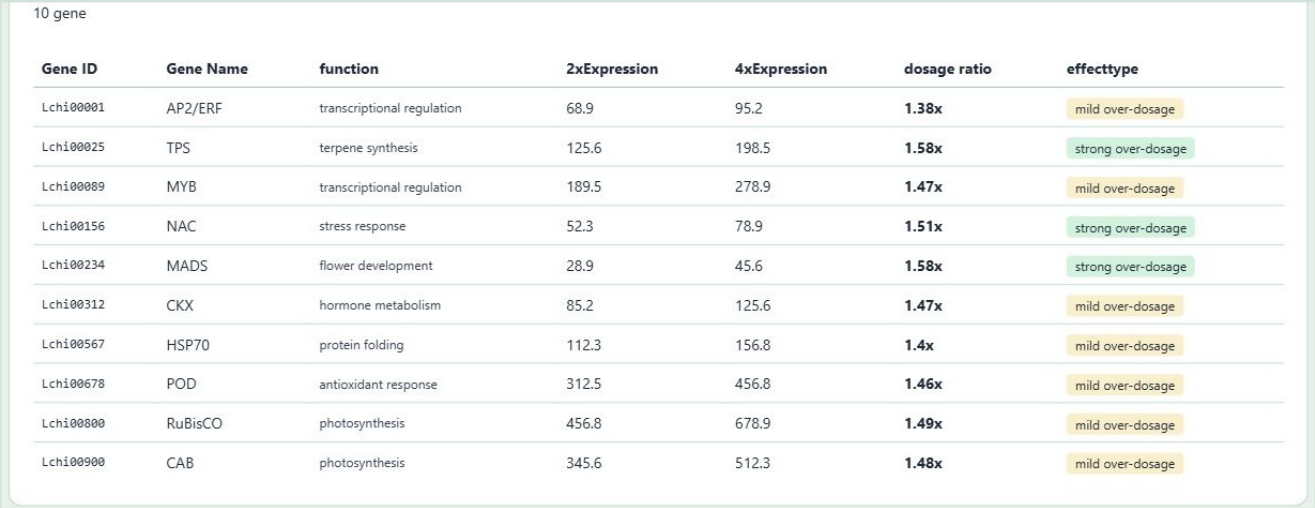
searches for genes in the diploid and tetraploid expression records included in the page, checks the expression ratio, and filters according to the existing threshold on the page.

2026-10-21 has been opened from the online address and taken a screenshot.



Ploidy Dosage·Ploidy dose record

real online screenshot: 2026-10-21.



10 gene

Gene ID	Gene Name	function	2xExpression	4xExpression	dosage ratio	effecttype
Lchi00001	AP2/ERF	transcriptional regulation	68.9	95.2	**1.38x**	mild over-dosage
Lchi00025	TPS	terpene synthesis	125.6	198.5	**1.58x**	strong over-dosage
Lchi00089	MYB	transcriptional regulation	189.5	278.9	**1.47x**	mild over-dosage
Lchi00156	NAC	stress response	52.3	78.9	**1.51x**	strong over-dosage
Lchi00234	MADS	flower development	28.9	45.6	**1.58x**	strong over-dosage
Lchi00312	CKX	hormone metabolism	85.2	125.6	**1.47x**	mild over-dosage
Lchi00567	HSP70	protein folding	112.3	156.8	**1.4x**	mild over-dosage
Lchi00678	POD	antioxidant response	312.5	456.8	**1.46x**	mild over-dosage
Lchi00800	RuBisCO	photosynthesis	456.8	678.9	**1.49x**	mild over-dosage
Lchi00900	CAB	photosynthesis	345.6	512.3	**1.48x**	mild over-dosage

Ploidy Dosage·Ploidy dose record

real online screenshot: 2026-10-21.

Input and parameters

- Search GenesID, nameorfunction...: Gene ID, name or function keyword.
- Group buttons: All, dosage compensation, normal dosage, mild over-dosage, strong over-dosage.

Sample process

Enter TPS → Click Lchi00025 → View 2x=125.6, 4x=198.5 and dosage ratio=1.58 recorded on the page → Download the filtered results.

Step by step operation

1. Enter Lchi00025, TPS or function keywords in the search box; no need to click the search button, the list changes instantly.
2. Click a group button to display only records belonging to that category and matching keywords; use All to restore all categories.
3. Click the pie chart icon to switch the display of the upper chart.
4. Click a row in the data table, or the gene mark in the dosage-ratio distribution track to expand the 2x/4x expression and ratio of the gene.
5. Click CSV, JSON or TSV to download the current filtered records.

How to read the result

- Table displays Gene ID, Gene Name, function, 2xExpression, 4xExpression, dosage ratio, effecttype.
- dosageRatio is the 4x/2x expression ratio recorded by the page. The page grouping thresholds are <0.9: dosage compensation; 0.9-1.1: normal dosage; >1.1-1.5: mild over-dosage; >1.5: strong over-dosage.
- Statistics cards and pie charts are based on all built-in records; searches and category filters affect tables, ratio bar charts, and downloads.

Frequently Asked Questions and Usage Restrictions

- This is a query and display of 10 records. There is no entry to upload the matrix or calculate the dose effect from the raw reads.
- The page grouping label is inconsistent with the description of the theoretical 2x expression at the top of the page; the manual only explains according to the current interface threshold, and the label cannot be directly interpreted as a statistically verified dose compensation or overdose conclusion.
- The page ratio is provided by data constants and is not recalculated from the expression value every time; the ratio is rounded.
- No biological replicates, significance tests and FDR results.

Transformation · Genetic transformation resources

breeding and ploidy | Data card and recipe resource browsing

Entrance

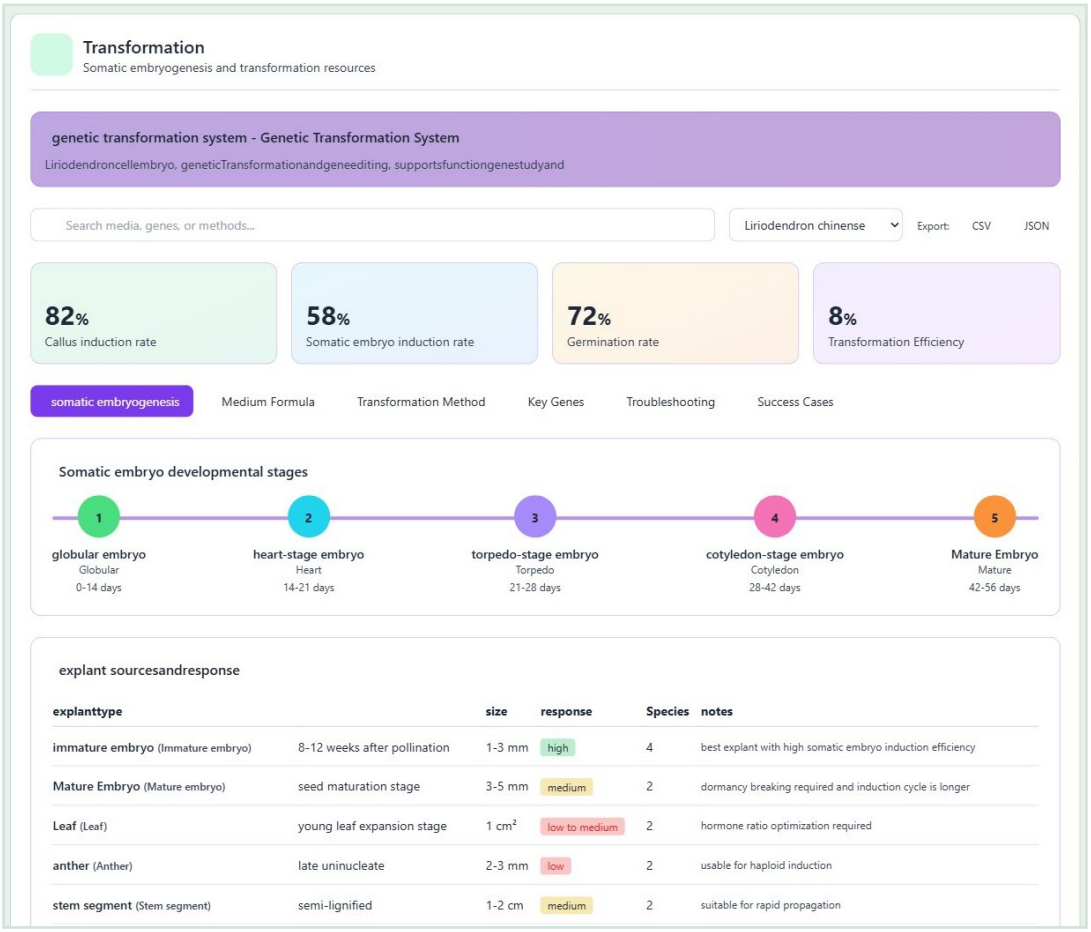
Analysis Tools → Breeding & Ploidy → Transformation

Open website/independent page

Function

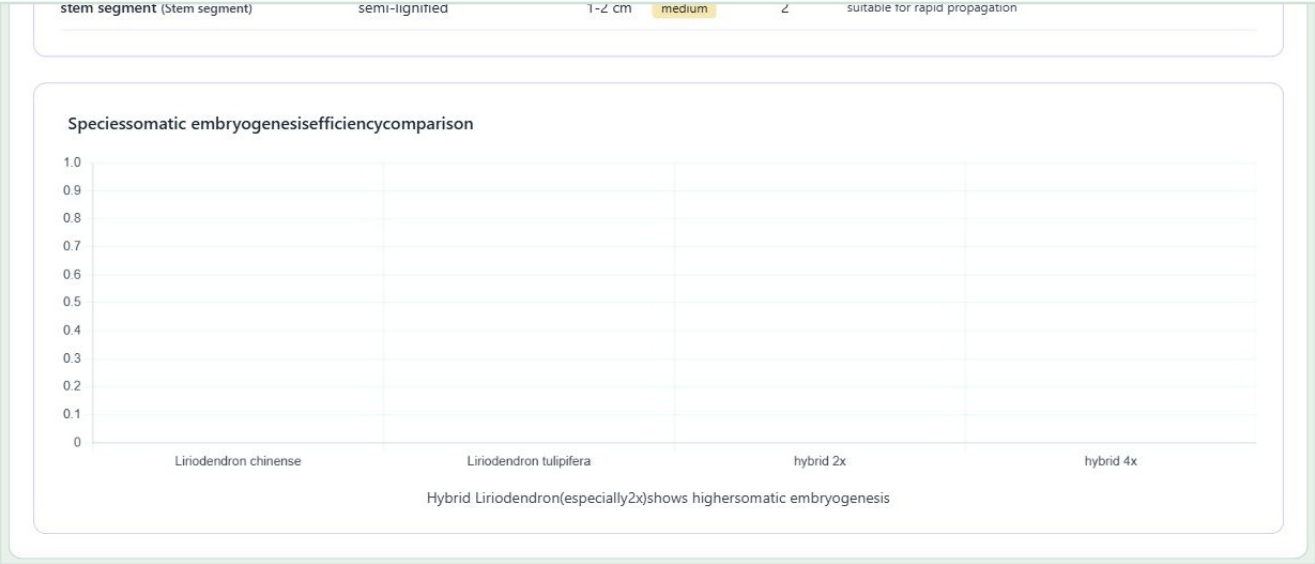
Browse resources such as somatic embryogenesis, culture medium formulations, transformation methods, key genes, FAQs and cases.

2026-10-21 has been opened from the online address and taken a screenshot.



Transformation · Genetic transformation resources

real online screenshot: 2026-10-21.



Transformation · Genetic transformation resources

real online screenshot: 2026-10-21.

Input and parameters

- Species drop-down box: Liriodendron chinense, Liriodendron tulipifera, Hybrid Liriodendron (2x), Hybrid Liriodendron (4x).
- Search box Search media, genes, or methods...: Actually only filters the culture medium name, stage and basic culture medium.
- Tag page: somatic embryogenesis, Medium Formula, Transformation Method, Key Genes, Troubleshooting, Success Cases.

Sample process

Enter Medium Formula → Search for callus → Read callus induction medium formula card → Download CSV; clear the search and switch to Transformation Method, click on the Agrobacterium method card to view the summary.

Step by step operation

1. Select the species to view the four percentage cards above: Callus induction rate, Somatic embryo induction rate, Germination rate, and Transformation Efficiency.
2. Enter somatic embryogenesis and click on the stage dots in the development timeline to view time, size, morphological characteristics and Marker genes; explants and species comparison charts are listed below.
3. Enter Medium Formula and enter MS, Induction or a fragment of the medium name in the search box; read the basic medium, sugar, agar, pH, hormone, cycle and remarks in the formula card.
4. Enter Transformation Method, click on the method card to expand the advantages, disadvantages and summary process; click again to collapse.
5. Enter Key Genes, Troubleshooting and Success Cases in order to read gene cards, problems and handling suggestions and cases.
6. Click CSV/JSON on the Medium Formula tab to download the current screening formula; click Download on other tabs to output Key Genes data.

How to read the result

- Species selection only switches four efficiency statistics cards; recipes, methods, genes and case lists are not filtered with species.
- This module provides information cards and summary processes, and does not perform actual genetic transformation, gene editing calculations, or experimental design optimization.
- The download file name transformation_media corresponds to the recipe; embryogenesis_genes corresponds to the key gene list.

Frequently Asked Questions and Usage Restrictions

- Although the search box says genes/methods, the current page only filters Medium Formula data; other tab pages will not respond to the search.
- The reference field contains Internal Data, under review, or just the journal abbreviation, which cannot be cited as a complete verifiable reference.
- Formulas, efficiencies and cases are built-in resource records; the corresponding species, genotypes, original literature and specific experimental conditions must be checked before use.
- There is no button to download the full experiment SOP.

Pathway · Pathway query

Metabolism and Traits | Preset pathway annotation query

Entrance

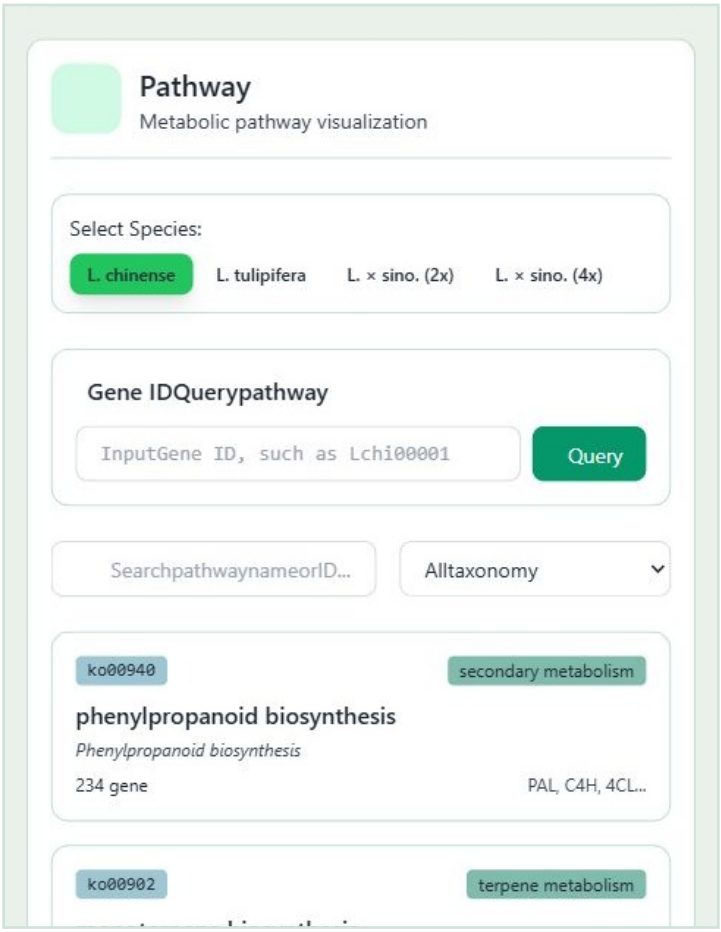
Analysis Tools → Metabolism & Traits → Pathway

Open website/independent page

Function

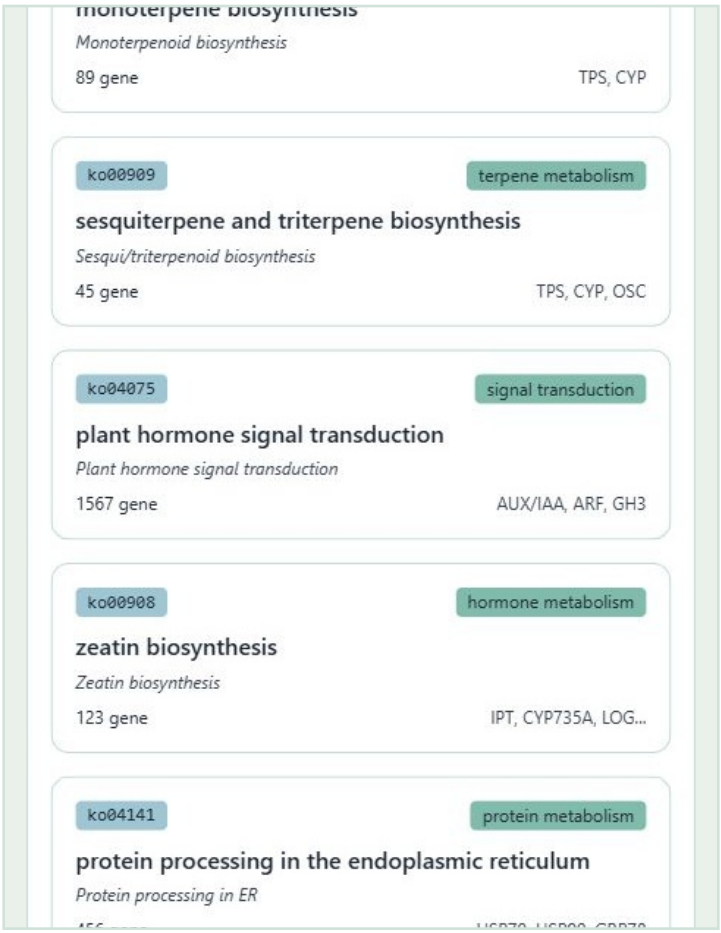
retrieves KEGG pathway annotations of built-in genes by species, browses a small number of pathway directories and jumps to KEGG pathway pages.

2026-10-21 has opened the online page and taken a screenshot. The narrow screen layout is retained in the figure, and the query and result areas are displayed in two sections.



Pathway · Pathway Query

real online screenshot: 2026-10-21.



Pathway · Pathway Query

real online screenshot: 2026-10-21.

Input and parameters

- Select Species: Select species/ploidy group first.
- Gene IDQuerypathway input box: A complete gene ID corresponding to the selected species.
- SearchpathwaynameorID...: Pathway name or ID keyword.
- Alltaxonomy: Channel category drop-down box.

Sample process

Select Species Select L. chinense → enter Lchi00025 → Query to view the terpenoid pathway annotation of the built-in gene; search below for ko00902 → click on the card → open KEGG.

Step by step operation

1. First use Select Species to select species; switching species will clear the expanded pathway and gene query results.
2. Enter the complete gene ID, such as Lchi00025 for Liriodendron tulipifera, into the Gene IDQuerypathway, click Query or press Enter.
3. Read the returned pathway ID, pathway name, KO and KO name; if it prompts not found, check the species, ID and current inclusion range.
4. Enter monoterpenoid or ko00902 in the separate pathway search box below; select Alltaxonomy or specify a category to filter cards.
5. Click on a pathway card to expand pathwayID, English, Gene Count, taxonomy, function, enzyme and some related genes.
6. Click inKEGGViewcompletepathway to open the KEGG page of the pathway in a new window.

How to read the result

- The upper part is the single gene precise ID table lookup; the lower part is the pathway catalog keyword filtering, and the two are independent query areas.
- pathwaygene(partial) in pathway details explicitly lists only partial genes.
- Gene Count is an existing record field. It is not a real-time statistics of this query, nor is it an enrichment significance result.

Frequently Asked Questions and Usage Restrictions

- Currently a small number of built-in annotations and pathway catalogs, no KEGG enrichment calculations of user gene sets are performed.
- Gene IDs must match exactly (including case); multiple IDs cannot be submitted in this input box.
- The module does not have a data download button; the external KEGG page needs to be connected to the Internet and used according to its page functions.
- No annotation hit only means that it is not included in the current built-in table. It cannot be inferred that the gene has no relevant biological function.

Floral Scent · Floral scent metabolism

Metabolism and Traits | Browse preset compounds and gene records

Entrance

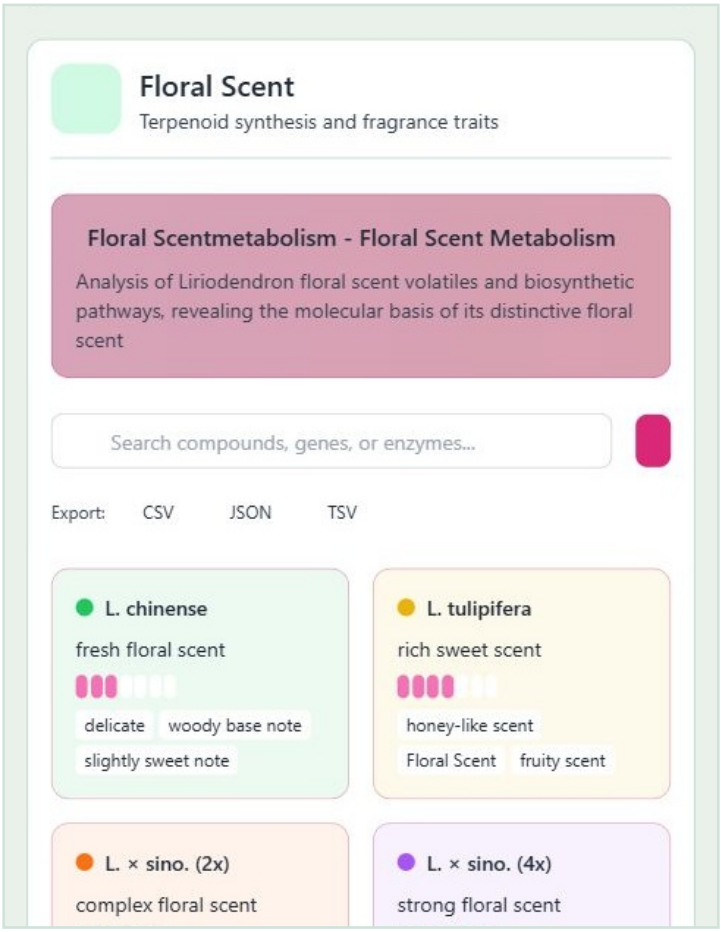
Analysis Tools → Metabolism & Traits → Floral Scent

Open website/independent page

Function

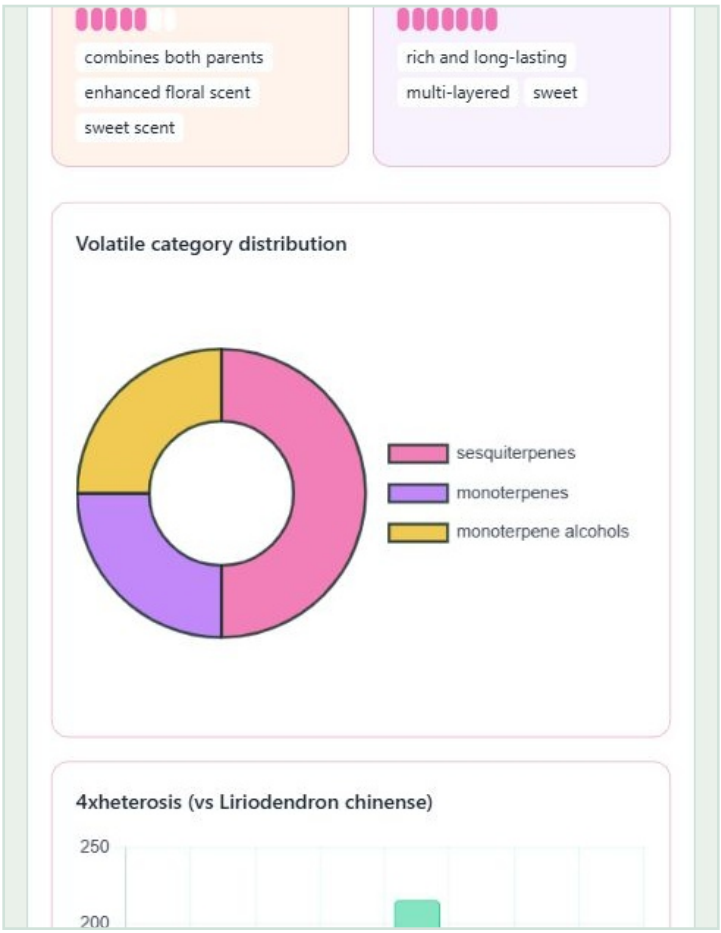
searches for built-in floral volatiles and synthetic genes, compares existing abundance records of four species/ploidy groups, and views simplified terpene synthesis diagrams.

2026-10-21 has opened the online page and taken a screenshot. The narrow screen layout is retained in the figure, and the query and result areas are displayed in two sections.



Floral Scent · Floral Scent Metabolism

real online screenshot: 2026-10-21.



Floral Scent · Floral Scent Metabolism

real online screenshot: 2026-10-21.

Input and parameters

- Search compounds, genes, or enzymes...: Volatile name/English name/category, or gene ID/name/enzyme name keyword.
- View buttons: Volatile composition, biosynthetic genes, Biosynthetic pathways.

Sample process

Volatile composition → Search linalool → Click this item → View Lchi00457/Lchi00458 → Cut biosynthetic genes → Clear and search for Lchi00458 → Export gene table.

Step by step operation

1. Select Volatile composition and enter linalool, myrcene or monoterpenes in the search box; the volatile comparison list will be filtered instantly.
2. Click on a volatile record to expand the compound name, category, aroma description, number of associated genes and gene ID.
3. Select biosynthetic genes, enter Lchi00458 or synthase, and view the corresponding genes, enzyme functions, catalytic reactions, and FlowerExpression(TPM).
4. Select Biosynthetic pathway and read the simplified route from pyruvate + G3P to GPP/FPP to volatiles.
5. Click the chart icon to show/hide the volatile category pie chart and the 4x abundance change plot relative to L. chinense.
6. Use CSV/JSON/TSV to export screened volatiles in the Volatile composition view; export screened genes in the biosynthetic genes view.

How to read the result

- Four aroma profile cards display the preset descriptions and intensities of Chinese Liriodendron, North American Liriodendron, and Hybrid 2x/4x respectively.
- The four colors of the volatile bar contrast correspond to the four species/ploidy groups; the exact values can be read in the abundance field of the exported file.
- The 4xheterosis (vs Liriodendron chinense) in the figure is actually (4x abundance/Chinese Liriodendron abundance-1)×100%, which is not a relative hybrid vigor of medium or high parent value.
- biosynthetic genes displays the expression values of floral tissues included in the page, and the path view is a simplified schematic diagram.

Frequently Asked Questions and Usage Restrictions

- Currently has 8 built-in volatiles and 8 synthetic genes, and does not support uploading raw mass spectra or performing metabolite identification.
- The search works on data in the current table; aroma description cards, category pie charts, and pathway representations are not reanalyzed with the search.
- Biosynthetic pathway view. Clicking to download still exports the gene table, not the pathway picture; there is no dedicated button to download the pathway diagram.
- Clicking on a gene table row will only highlight it, and there is currently no additional gene details panel.
- Abundance units, sampling conditions, replicates, and statistical tests are not fully specified; formal conclusions should be reviewed back to the original study.

AI Assistant · Database Q&A

AI Assistant | External Language Model Q&A

Entrance

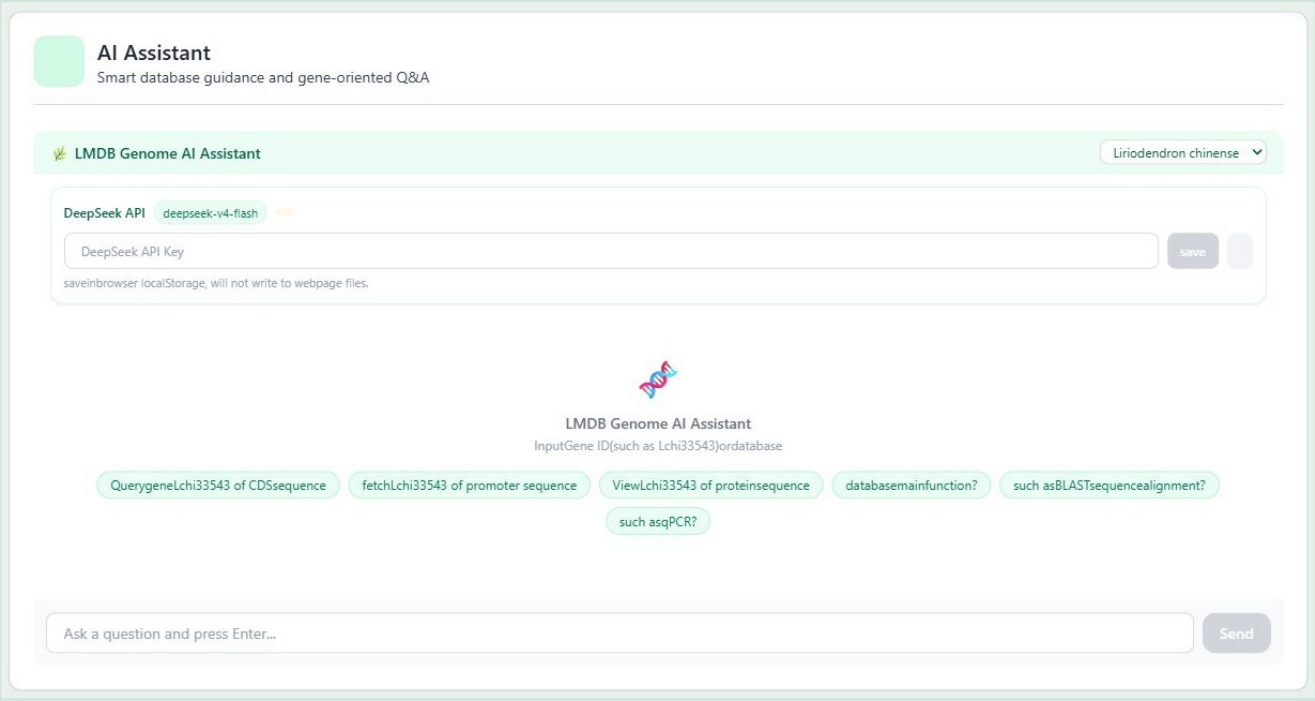
Analysis Tools → AI Assistant → AI Assistant; the AI Assistant floating entrance on the lower right side of the homepage also uses the same assistant.

Open website/independent page

Function

obtains database entry instructions, operation guidance and gene-related questions and answers through DeepSeek dialogue; whether the real gene sequence can be read depends on the separately configured genome service.

2026-10-21 has opened the online interface and taken a screenshot. The API Key has not been filled in and the conversation request has not been submitted; the AI answers and genome data connections have not yet been verified in this round.



AI Assistant · Database Q&A

real online screenshot: 2026-10-21.

Input and parameters

- A valid personal DeepSeek API Key; fill it in through the assistant's DeepSeek API password input box and click save. The key is saved to the browser's localStorage for the current website on this machine.
- Select Liriodendron chinense or Liriodendron tulipifera; enter a natural language question such as How to perform BLAST search?.
- If the administrator has connected to the genome service, you can enter the Chinese Liriodendron tulipifera gene ID such as Lchi33543 or the North American Liriodendron tulipifera ID in Litul. format supported by the service, and specify CDS, protein or promoter.

Sample process

Fill in the key and save → Enter How to perform BLAST search? → Send. If the interface shows Genome library connected, try Get CDS of Lchi33543 again and check the returned genes and species.

Step by step operation

1. Open AI Assistant and select the target species in the top species drop-down box.

2. Fill in your valid API Key in the DeepSeek API area and click save. This assistant currently calls DeepSeek directly; if the key is not filled in, it will prompt requires DeepSeek API Key.
3. At the bottom Ask a question and press Enter... Enter the question, press Enter or click Send. You can also click on the page to preset questions.
4. Wait for Thinking... to complete and check the answer in the dialogue area. You can continue to ask the previous question.
5. When a sequence is required, first confirm whether Genome library connected appears. Request CDS, protein or promoter by species and correct gene ID only if the administrator has configured the service and the status is connected.
6. If a request error is encountered, check the key validity, service access conditions and network; when the genome service is not connected, the actual downloaded data should be used to check the sequence.

How to read the result

- Text answers in chat form, which can be used for database function description and operation guidance.
- Only when the configured genome service successfully returns data, the answer context will contain real data such as gene coordinates, CDS, proteins, promoters, etc.; some long sequences may be truncated by the service.

Frequently Asked Questions and Usage Restrictions

- The current assistant requires a valid personal DeepSeek API Key and relies on external services and networks. The existence of backend files or the display of AI portals on the site does not mean that the current assistant has access to available public agents.
- The assistant can only read the corresponding sequence if the administrator configures it separately and successfully connects to the genome service. When there is no Genome library connected status, do not take connection statements in preset questions or answers as proof that real data has been read.
- Genetic data extraction handles up to the first 3 unique IDs in the problem at a time; currently recognizes Lchi numeric and Litul alphanumeric formats. Other formats need to confirm service support first.
- This is a conversational guide and does not automatically run BLAST, primer design, statistical enrichment, or WGCNA for the user. Answers involving sequence, function, and analytical conclusions should be checked against the original data.
- There is currently no chat record file export button; if the key or service request fails, check the page prompts and configuration first.

Gene Structure Viewer · Gene structure visualization

additional page | existing annotation structure query

Entrance

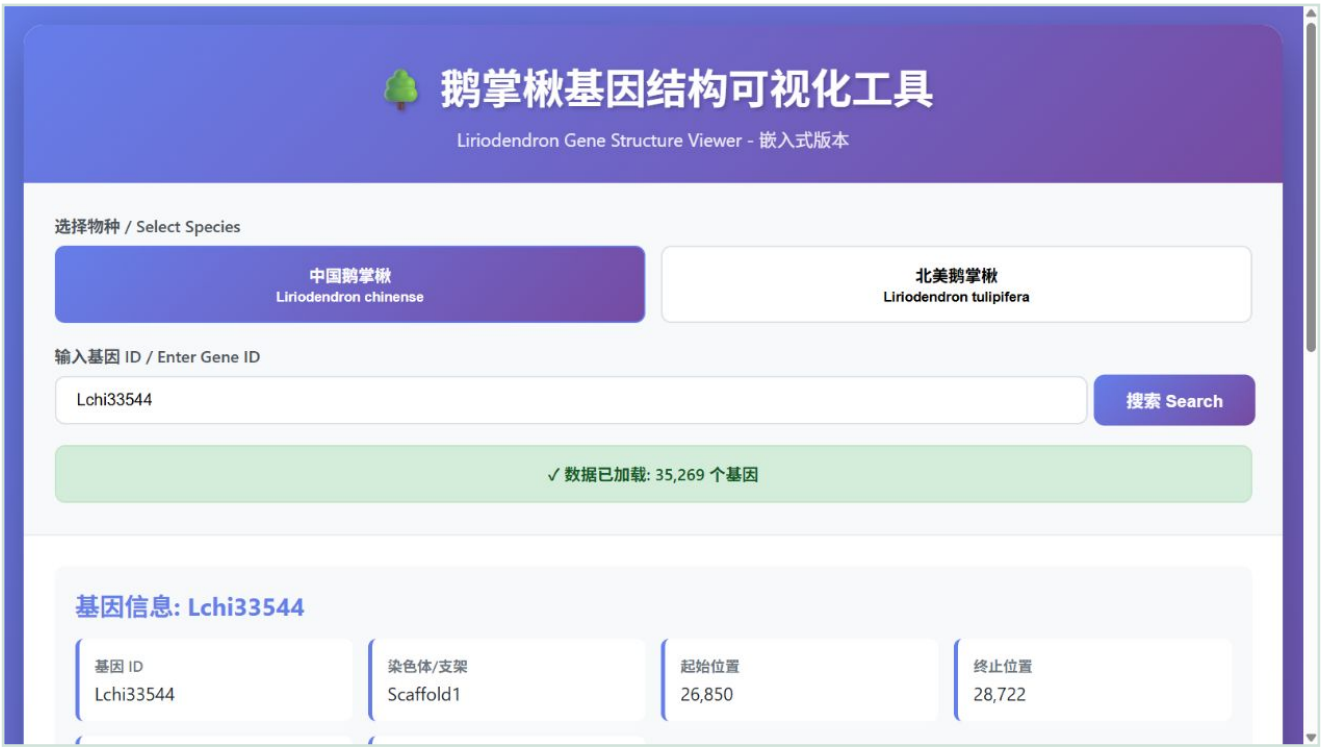
This independent tool is not listed in the current main menu. Please click the independent page link below to enter the Gene Structure Viewer.

Open website/independent page

Function

View the chromosome/scaffold position, strand direction, exon/CDS number and structure diagram based on the Chinese or North American Liriodendron tulipifera gene ID. This page is independent of the main tool Genome Browser.

2026-10-21 The online page shows that 35,269 Chinese tulipifera genes have been loaded. Lchi33544 has been entered and queried, and the gene information and location can be seen in the screenshot; the complete structure diagram below has not been screenshotted.



Gene Structure Viewer · Gene structure visualization

real online screenshot: 2026-10-21.

Input and parameters

- Click on Liriodendron chinense or Liriodendron tulipifera in Select Species.
- Enter a complete gene ID in Enter Gene ID, such as Lchi33544 or Litul.01G000100. Supports case-insensitive searches.

Sample process

selects Liriodendron tulipifera → enter Lchi33544 → Search Search → view the structure diagram; then select Liriodendron tulipifera North America → enter Litul.01G000100 → Search Search. Both examples exist in the same directory, gene_data.js.

Step by step operation

1. Open the independent page, and the waiting status shows "□ Data loaded". Liriodendron tulipifera is selected by default.

2. Click the target species button; Chinese Liriodendron tulipifera displays 35,269 genes, and North American Liriodendron tulipifera displays 29,655 genes (currently included).
3. Enter the complete gene ID, such as Liriodendron tulipifera Lchi33544, click Search or press Enter.
4. View the gene ID, chromosome/scaffold, start position, end position, strand direction, gene length, and exon number and CDS number in Gene Information.
5. Look down at the structure diagram: green is Exon, blue is CDS, orange is 5′ UTR, red is 3′ UTR, gray lines indicate gene mainline/intron segments; gray arrows indicate positive and negative strand directions, and the numbers at the bottom are genome coordinates.
6. After switching to Liriodendron tulipifera, re-enter Litul.01G000100 and search; switching species will hide the previous result.

How to read the result

-Basic position and structural quantity information of a hit gene. Gene lengths are shown in bp by end-start+1. The

- Canvas structure diagram is arranged from left to right according to genome coordinates, showing the Exon, CDS and existing UTR segments of the gene in the data. The negative strand is marked with a left-pointing arrow.

Frequently Asked Questions and Usage Restrictions

- This is an independent page, used separately from the main menu Genome Browser.
- Must be deployed together with gene_data.js in the same directory. If the file is not uploaded, the path does not match, or the loading fails, the page will prompt that the genetic data is not loaded and cannot be queried.
- Only one gene is displayed at a time; the implementation uses find to find the first record containing the input string, and the accuracy and inclusion judgments are written in the same condition. To avoid short words hitting other records first, enter the full ID and check the resulting title.
- No batch comparisons, transcript switching, GFF/GTF uploads, sequence extraction, zoom pan, mouseover details or download buttons.
- When the Exon/CDS/UTR information is missing in the data, it will not be compensated; an empty structure does not mean that the corresponding biological structure does not exist.

Comparative Genomics DotPlot · Whole-genome comparison

additional page | Precomputed genome alignment map

Entrance

Omics Data → Genomics → Comparative Genomics; upper right of the page Open opens the same comparative_genomics/ctg.html in a new window. Independent website address: https://awzh15252056628.github.io/liriodendron-multiomics-database/comparative_genomics/ctg.html. This is a separate comparison page under the genomic data area, and is not the same page as Analysis Tools → Synteny Analysis.

Open website/independent page

Function

interactively browses the existing aligned fragments of the whole genome of Liriodendron tulipifera from China and Liriodendron tulipifera from North America, zooming, panning and viewing the fragment coordinates.

This section organizes operations based on existing page files. Since this round of browser access was set to be closed, this independent online address was not verified, and no real-time screenshot of the page was obtained. This status does not indicate a website failure; actual controls and online availability are subject to the page after opening.

Input and parameters

- No need to enter gene ID or upload files; comparison map and coordinate data are already built into the page.
- The input operation is a mouse frame selection, moving graphics or hovering a fragment; the specific name of the toolbar and available buttons are subject to the browser display.

Sample process

Enter Comparative Genomics → Open → Select the intersection area between LCHI_Chr000001 and LTUL_Chr000001 → Hover the line segment to view the coordinates on both sides → Double-click to restore the full image.

Step by step operation

1. Select Genomics in Omics Data, then click the Comparative Genomics card and wait for DotPlot to display; click Open when wider space is needed.
2. Read the coordinate axes: the horizontal axis Target is LCHI_Chr... (Chinese Liriodendron tulipifera), and the vertical axis Query is LTUL_Chr... (North American Liriodendron tulipifera). The chromosomes in the figure are arranged according to the current alignment layout, not necessarily in numerical order.
3. Move the mouse into the picture, select Zoom on the upper right toolbar, and drag out a rectangular frame to enlarge the chromosome crossover area of interest.
4. Select Pan to pan the graph, and then drag to view adjacent areas; click Reset axes/Autoscale on the toolbar or double-click in the graph to restore the field of view. The current page implementation explicitly configures double-click as reset.
5. Hover the mouse over the comparison line segment to read Query ID, Query Start Pos, Query End Pos, Target ID, Target Start Pos, Target End Pos and Length.
6. If the camera icon Download plot as a png appears in the toolbar, you can click to save the current graph; this item belongs to the Plotly graphics toolbar and cannot replace the original comparison coordinate file download.

How to read the result

- Chromosome-level interaction lattice/fragment plot of Liriodendron tulipifera Target from China and Liriodendron tulipifera Query from North America.
- Built-in figure title record: Post-filtering number of alignments: 2212; minimum alignment length (-m): 10000; Post-filtering number of queries: 19; minimum query aggregate alignment length (-q): 4e+05. They are fixed settings and results when generating graphs.
- The hover prompt displays the chromosomes on both sides of the aligned fragment, the start and end coordinates, and the length.

Frequently Asked Questions and Usage Restrictions

- The graph is a pre-calculated result; the web page cannot upload new genomes, re-execute minimap2, change the filtering threshold or re-call collinearity analysis.
- There is no input box to search by gene ID, nor a selector to switch to hybrid diploid/tetraploid.
- The current page implements embedded graphics scripts and data, and does not rely on the lchi_vs_ltu.paf.gz next to it when running; the existence of this PAF file in the warehouse does not mean that the original result download button is provided on the picture.
- Use the toolbar Zoom box to zoom. Although the homepage prompts Scroll to zoom, the embedded image does not explicitly enable scroll wheel zoom. The scroll wheel operation is subject to the actual page performance.
- This page does not provide SV event classification, statistical significance, Ka/Ks, or gene family expansion and contraction calculations. Extend describes the expandable direction.

Tetraploid Dosage Effect Analyzer · Tetraploid dosage results viewer

additional page | Drawing after the user imports the precalculated results table

Entrance

The current index menu does not link to this page; direct link: Tetraploid_Dosage_Effect_Analyzer.html.

Open website/independent page

Function

loads the calculated full_results.csv, interactively browses dosage categories, differential expression plots, functional statistics, heat maps and single gene expression, and exports charts and filtering results.

This section organizes operations based on existing page files. Since this round of browser access was set to be closed, this independent online address was not verified, and no real-time screenshot of the page was obtained. This status does not indicate a website failure; actual controls and online availability are subject to the page after opening.

Input and parameters

- Choose CSV File: UTF-8 encoded, comma-delimited full_results.csv; supports file suffix .csv/.txt.
- Fields: gene_id, D_mean, T_mean, D_std, T_std, log2_ratio, fold_change, pvalue, padj, neg_log10_p, category, significant, functional_category, gene_description.
- Fixed sample columns: D_1_1_fpkm, D_1_2_fpkm, D_1_3_fpkm, D_2_1_fpkm, D_2_2_fpkm, D_2_3_fpkm, D_3_1_fpkm, D_3_2_fpkm, D_3_3_fpkm, and corresponding There are 9 columns in total from T_1_1_fpkm to T_3_3_fpkm.

-significant should use case-accurate True/False; without these precomputed fields, the page will not do the difference check.

Sample process

loads its own full_results.csv → Data Table input the complete gene_id already in the file → select Significant only → click the results row to view the sidebar → Download Filtered CSV.

Step by step operation

1. Open the page through an independent address, click Choose CSV File to load the prepared full_results.csv; if the existing file is .csv.gz, decompress it first and then select it.
2. Overview View the total number of genes, number of significant genes, dose category statistics and preset category explanations; the Export Pie/Export Bar next to the picture can export SVG.
3. Volcano Plot Use Color by to switch dosage categories, functional categories or significance; you can set p-value threshold, |log□FC| and Max points, hover to view genes, and click points to open the single-gene sidebar.
4. MA Plot View the relationship between the expression mean and log□(T/D), set the color and Max points; click on the points to view gene details.
5. Expression Enter at least 3 characters to get gene suggestions, click the suggestion or enter the complete and accurate ID and click Show Expression; view the sidebar D/T column plot, box plot, mean, p/FDR and comments for each sample.
6. Functional View the number of functional categories, dose × function heat map and average log□FC for each functional category; use All dosage categories to filter the quantity map.
7. Heatmap settings Category, Top N genes, Sort by and Scale. Optional Z-score (row), log2(FPKM+1) or Raw FPKM.
8. Data Table Enter gene ID or description keywords, filter by category, function and Significant only/Non-significant combination; click the header to sort, Next/Prev to turn pages, and click on the row to open the sidebar.
9. Export or the button next to the figure can save the chart; Download Filtered CSV downloads the current filtered collection of the data table, Significant Genes CSV exports the records with significant=True in the full data, and Summary CSV downloads the category summary.

How to read the result

- D=diploid, T=tetraploid; log□(T/D)>0 means the T/D ratio is greater than 1, <0 means less than 1.
- All pvalue, padj, significant and category are from the input CSV, the page does not calculate these statistics.
- Heatmap Z-score is normalized gene-by-gene across 18 fixed sample columns, indicating the relative change of the gene between samples; Raw FPKM is used to read the raw values.
- Volcano plot threshold control only moves the reference line; significance coloring with Significant Genes CSV still uses the significant field within the CSV.
- Show Expression Open the gene panel on the right; the gene chart should be read in the side column and do not wait for another result to be generated in the main image area.

Frequently Asked Questions and Usage Restrictions

- The page only loads pre-calculated results and does not accept FASTQ/counts to directly calculate differences, p-values or dose groupings; the display of charts after file upload cannot be regarded as re-completed statistical analysis.
- The heat map and gene expression map are programmed with 9 D and 9 T sample columns; missing columns will be drawn as 0, and the format should be matched first.
- Input must provide neg_log10_p, volcano plot will not be automatically complemented by pvalue.
- When filtering is empty or there are no significant genes, the CSV download may fail; first confirm that the result is not empty.
- Its dosage_compensation is defined as T/D≈1, which is different from the label threshold of the main menu Ploidy Dosage, and the two page classifications should not be mixed.

Terminology and result judgment

ID and accession

gene ID corresponds to the specific species and annotation version; BioProject, SRA run, and MassIVE accession correspond to the project or public archive record. Record the complete number.

FASTA / CDS / protein

FASTA is the sequence format with ">name" header line; CDS is the coding DNA, and protein is the amino acid sequence. Different tools have different requirements, so do not just replace the sequence with the gene name.

FPKM / TPM

expression unit, used to view the provided expression records. Higher values do not mean significant differences; cross-dataset comparisons also require a combination of samples, processing, and normalization.

P value / padj / FDR

padj is the corrected P value field. FDR requires corresponding statistical tests and multiple corrections; tools that only have annotation lists or expression diagrams do not provide such tests.

identity / E-value

identity is the proportion of identities within aligned fragments, and E-value is the expected number of such matches to be obtained by chance. Also look at coverage, species, sequence quality and alignment conditions.

Tm / GC% / PAM

Tm is the estimated annealing-related temperature, GC% is the base composition ratio, and PAM is the adjacent sequence required for Cas recognition. They are not a substitute for primer specificity or genome-wide off-target checking.

pLDDT

Residue-by-residue confidence score for the predicted structure; only models that have explicitly saved this field as confidence can be interpreted this way. The B-factor of an experimental PDB usually has another meaning.

Ho / F / kinship / Fst

is used to observe heterozygosity, inbreeding correlation statistics, kinship and population differentiation respectively. The results of group page views on this site have been calculated, and these indicators are not recalculated by uploading VCF.

MPH / HPH / 4x:2x

MPH compares the average value of the medium parent, and HPH compares the high parent value; 4x:2x is the ratio of tetraploid to diploid. The classification definitions of the main page and the independent dosage tool are different and should be read according to the corresponding chapters.

Precomputation / Example / External Service

precalculation is a query or display of existing results. An example is limited records on the page. External services require online submission or jump. These modes each have different input and output ranges.

Download, record and troubleshoot

No results found in search

First clear the superposition conditions such as species/category and use the page example to confirm the format. Gene query checks case and complete ID, and project search uses complete accession; no record in the limited directory does not mean that the biological object does not exist.

The page keeps loading

Wait for data to be loaded and confirm that the browser and network are normal; the expression page must support modern decompression API. The independent structure page needs to be accompanied by annotation data files, and the chart page needs its visualization dependencies to be loaded normally.

External request failed

records the error and task ID. BLAST retains the RID and checks it at NCBI; UniProt, ESMFold and AI rely on external services. AI also requires a valid key or a configured proxy as required by the page.

The exported file is not original data

differentiates between catalog TSV/JSON, node or annotation tables, prediction PDBs, graph files, and raw sequencing/mass spectrometry files. Catalog export does not automatically package all raw data.

The image is empty after the user file is uploaded

Check the file name, encoding, delimiter, field, sample column and data level according to the corresponding chapter. The independent dose tool requires already calculated p-values and categories, and missing fields will not be automatically calculated by the web page.

Save and reference

records the database URL, access date, tool name, species/version, gene or project ID, input file, parameters, filter conditions and export file. Public data should also cite original projects and papers and comply with their licenses.

Recommended record template to save

Task: ____ | Date: ____ | Tool: ____

species / Annotated version: ____ | Input genes / Project: ____

Input files and parameters: ____ | Results and export: ____ | Original source: ____

Tool Index

Click a tool name to open its chapter. Page numbers refer to this English edition.

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